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Extremely Long Lived Localized Singlet Diradicals in a Macrocyclic Structure: A Case Study on the Stretch Effect

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Extremely Long-lived Localized Singlet Diradicals in a Macrocyclic Structure: A Case Study of Stretch Effect

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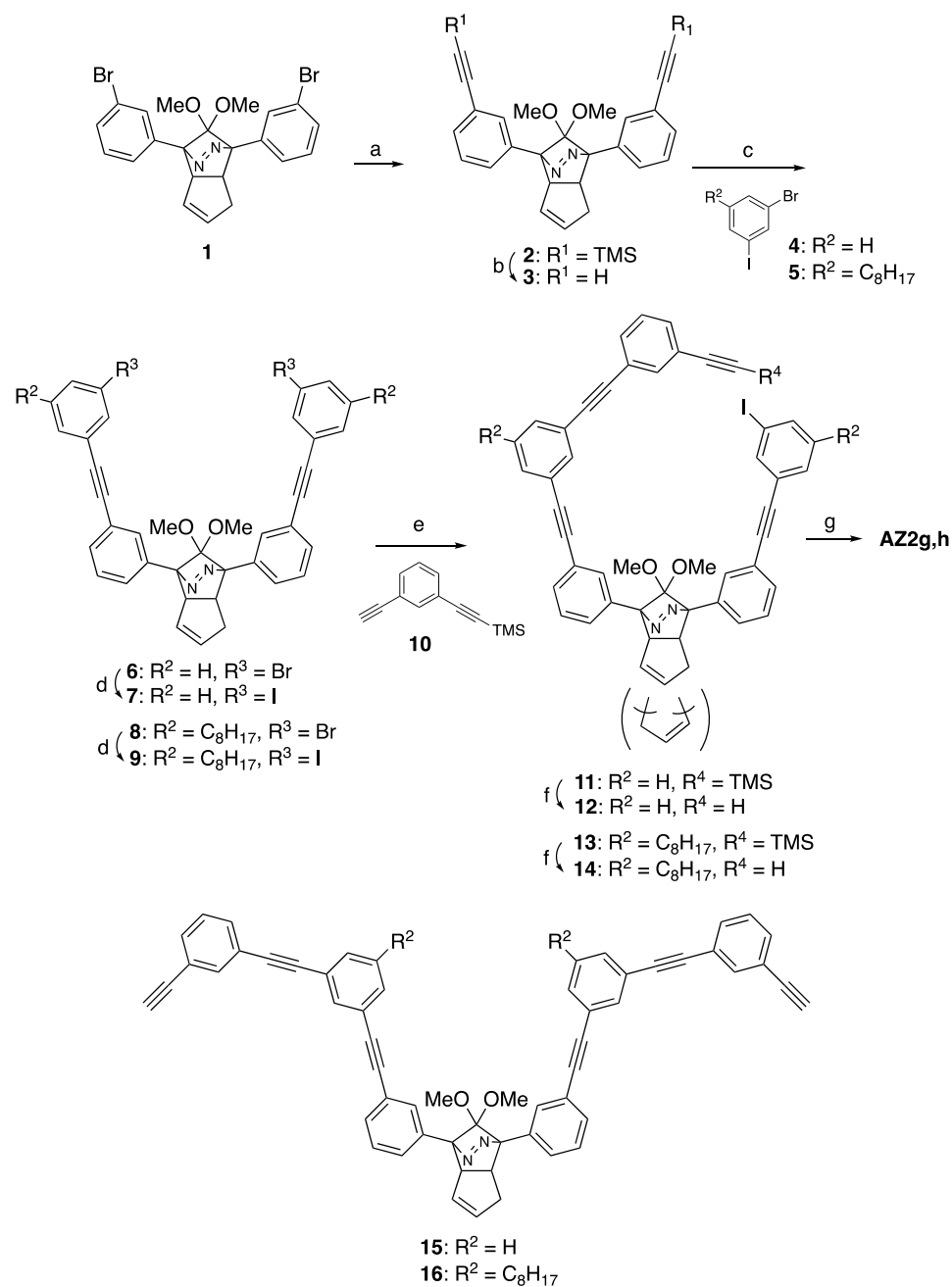
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Scheme S1. Synthesis of azo compound **1**. Reaction conditions: (a) NaH, THF, 0 °C $\rightarrow$ r.t., 24 h; (b) Ph₂Se₂, S₂O₈(NH₄)₂, MeOH, reflux, 2.5 h; (c) N₂H₄H₂O, CHCl₃, reflux, 13 h; (d) cyclopentadiene, TFA, CH₂Cl₂, 0 °C, 1 h. Synthesis of azoalkane **AZ2g,h**. Reaction conditions: (e) trimethylsilylacetylene, Pd(PPh₃)₄, CuI, THF, Et₃N, reflux, 18.5 h; (f) K₂CO₃, THF, MeOH, r.t., 4 h; (g) Pd(PPh₃)₄, CuI, THF, Et₃N, reflux, 19 h; (h) *n*-BuLi, I₂, THF, -78 °C, 5 h; (i)

Pd(PPh₃)₄, CuI, THF, Et₃N, reflux, 18.5 h; (j) K₂CO₃, THF, MeOH, r.t., 5 h; (k) Pd(PPh₃)₄, THF, Et₃N, reflux, 4 h.

Synthetic procedure.

(a) A solution of sodium hydride (2.4891 g, 62.2 mmol) in dried THF (40 mL) was stirred at room temperature for 30 min under nitrogen atm. Methyl 3-bromobenzoate (6.5308 g, 30.8 mmol) in THF (40 mL) and 3-bromoacetophenone (4.1 mL, 30.8 mmol) was added and stirred at 0 °C for 5 h, at room temperature for 15 h. The reaction mixture was added to 1 M HCl in crash ice and filtered to afford **1,3-bis(3-bromophenyl)propane-1,3-dione** as a white solid (10.3453 g, 27.1 mmol, 88 %). ¹H NMR (CDCl₃, 500 MHz): δ 8.11 (s, 2H), 7.92 (d, *J* = 7.77 Hz, 2H), 7.70 (d, *J* = 7.84 Hz, 2H), 7.38 (dd, *J* = 7.77 Hz, *J* = 7.84 Hz, 2H), 6.76 (s, 1H). ¹³C NMR (100 MHz, CDCl₃): δ 184.41 (C), 137.26 (C), 135.49 (CH), 130.29 (CH), 130.25 (CH), 125.78 (CH), 123.02 (C), 93.44 (CH). HRMS (ESI) Calcd for C₁₅H₁₀O₂Br₂Na [M+Na]⁺ = 402.89398, Found 402.89444.

(b) A solution of **1,3-bis(3-bromophenyl)propane-1,3-dione** (18.1933 g, 47.6 mmol), diphenyl diselenide (7.4263 g, 23.8 mmol) and ammonium peroxodisulfate (21.7230 g, 95.2 mmol) in MeOH (200 mL) was refluxed at 75 °C under nitrogen atm for 2.5 h. After 2.5 h, the water was added into the reaction mixture until the ammonium peroxodisulfate was dissolved. The organic product was extracted with chloroform, dried over MgSO₄ and filtered. After filtration, the solvents were removed to afford **1,3-bis(3-bromophenyl)-2,2-dimethylpropane-1,3-dione** as a red liquid. The crude product was directly used in the next step because **1,3-bis(3-bromophenyl)-2,2-dimethylpropane-1,3-dione** was not stable in silica.

(c) A solution of **1,3-bis(3-bromophenyl)-2,2-dimethylpropane-1,3-dione** (25.5858 g, 57.9 mmol) and hydrazine mono-hydrate (8.7 mL, 226 mmol) in CHCl₃ (200 mL) was refluxed at 70 °C for 11.5 h under nitrogen atm. After 11.5 h, 1M HCl was added and stirred for 30 min, then NaHCO₃ was added and the organic product was extracted with CHCl₃, washed with brine, dried over MgSO₄, and filtered. After filtration, the solvents were removed, and the residue was separated by column chromatography on silica gel with a solvent mixture (hexane/EtOAc 4/1) to afford **3,5-bis(3-bromophenyl)-4,4-dimethoxy-4H-pyrazole** as a yellow solid (3.9185 g, 8.9

mmol, 19 % (2step)). ^1H NMR (CDCl_3 , 400 MHz): δ 8.41 (s, 2H), 8.2 (d, $J = 8.0$ Hz, 2H), 7.7 (d, $J = 8.0$ Hz, 2H), 7.4 (dd, $J = 8.0$, $J = 8.0$, 2H), 3.07 (s, 6H). ^{13}C NMR (100 MHz, CDCl_3): δ 52.12, 117.09, 123.27, 126.33, 129.35, 150.51, 130.64, 135.49, 166.07. HRMS (ESI) Calcd for $\text{C}_{17}\text{H}_{15}\text{N}_2\text{O}_2\text{Br}_2$ $[\text{M}+\text{H}]^+ = 436.94948$, Found 436.95020.

(d) A two-necked round bottom flask was charged with **3,5-bis(3-bromophenyl)-4,4-dimethoxy-4H-pyrazole** (3.0230 g, 6.9 mmol) and cyclopentadiene (11.5 mL, 137 mmol) in DCM (100 mL) under nitrogen atm. Trifluoroacetic acid (0.1 mL, 1.3 mmol) was added and stirred at 0 °C for 1 h under nitrogen atm in the absence of light. After 1 h, sodium bicarbonate was added and the organic product was extracted with DCM, washed with brine, dried over MgSO_4 , filtered and concentrated. The residue was recrystallized with DCM/Hexane to afford **1,4-bis(3-bromophenyl)-8,8-dimethoxy-4,4a,5,7a-tetrahydro-1H-1,4-methanocyclopenta[d]pyridazine (1)** as a white solid (2.9410 g, 5.8 mmol, 84 %). ^1H NMR (CDCl_3 , 400 MHz): δ 8.13, 8.04 (dd, $J = 8.0$ Hz, 2H), 7.92, 7.83 (d, $J = 8.0$ Hz 2H), 7.54 (t, $J = 7.5$ Hz, 2H), 7.34 (ddd, $J = 16$, 4.0 Hz, 2H), 5.6-5.58, 5.49-5.47 (m, 2H), 4.12 (m, 1H), 3.67-3.66 (m, 1H), 3.00 (s, 3H), 2.71 (s, 3H), 2.34-2.18 (m, 2H). ^{13}C NMR (100 MHz, CDCl_3): δ 32.16, 42.40, 51.99, 52.34, 57.26, 92.24, 94.00, 118.39, 122.73, 122.80, 126.12, 126.55, 127.27, 130.02, 130.07, 131.04, 131.18, 131.33, 131.79, 134.60, 138.25, 138.43. HRMS (ESI) Calcd for $\text{C}_{22}\text{H}_{20}\text{O}_2\text{N}_2\text{Br}_2\text{Na}$ $[\text{M}+\text{Na}]^+ = 524.97837$, Found 524.97852.

(e) A two-necked round bottom flask was charged with **1** (2.80203 g, 5.6 mmol), tetrakis (triphenylphosphine) palladium (0.6504 g, 0.56 mmol), and copper iodide (0.1602 g, 0.84 mmol) under nitrogen atm. THF (80 mL), triethylamine (20 mL), and trimethylsilylacetylene (7.8 mL, 56 mmol) were added and refluxed at 70 °C for 18.5 h under nitrogen atm in the absence of light. After 18.5 h, the reaction mixture was filtered by celite and concentrated. The organic product was extracted with CHCl_3 , washed by aqueous NH_4Cl and brine, dried over MgSO_4 , and filtered. After filtration, the solvents were removed, and the residue was separated by column chromatography on silica gel with a solvent mixture (hexane/EtOAc 8/1) to afford **2** as a mixture (3.3773 g), hence it was directly used in the next step.

(f) A solution of **2** as a mixture (3.3244 g), K_2CO_3 (0.4260 g, 3.1 mmol) in THF (50 mL) and MeOH (80 mL) was stirred at room temperature for 4 h. After filtration by celite and

concentration, water was added. The organic product was extracted with EtOAc, washed with brine, dried over MgSO₄, filtered and concentrated. The residue was recrystallized to afford **1,4-bis(3-ethynylphenyl)-8,8-dimethoxy-4,4a,5,7a-tetrahydro-1H-1,4-methanocyclopenta[d]pyridazine (3)** as a white solid (1.9021 g, 4.8 mmol, 86 % (2step)). ¹H NMR (CDCl₃, 400 MHz): δ 8.10 (m, 1H), 8.01-7.99 (m, 2H), 7.92-7.90 (m, 1H), 7.54 (t, *J* = 8.0 Hz, 2H), 7.44 (ddd, *J* = 16, 4.0 Hz, 2H), 5.59-5.57, 5.50-5.48 (m, 2H), 4.17-4.14 (m, 1H), 3.72-3.67 (m, 1H), 3.12 (s, 2H), 3.00 (s, 3H), 2.70 (s, 3H), 2.40-2.17 (m, 2H). ¹³C NMR (100 MHz, CDCl₃): δ 32.18, 42.28, 51.90, 51.93, 52.24, 52.29, 57.12, 83.69, 83.71, 92.35, 94.08, 118.33, 122.30, 122.35, 126.27, 128.49, 128.53, 128.64, 129.42, 131.52, 131.71, 131.74, 131.87, 132.23, 134.49, 136.35, 136.53. HRMS (ESI) Calcd for C₂₆H₂₂O₂N₂Na [M+Na]⁺ = 417.15735, Found 417.15735

(g) A side-arm flask was charged with **3** (1.2020 g, 3.0 mmol), tetrakis (triphenylphosphine) palladium (0.3468 g, 0.30 mmol), and copper iodide (0.0861 g, 0.45 mmol) under nitrogen atm. 1-Bromo-3-iodobenzene (**4**) (1.1 ml, 8.7 mmol), THF (60 mL) and triethylamine (20 mL) were added and refluxed at 70 °C for 10 h under nitrogen atm. After filtration by celite and concentration, then water was added and the organic product was extracted with CHCl₃, washed by aqueous NH₄Cl and brine, dried over MgSO₄, filtered and concentrated. The residue was separated by column chromatography on silica gel with a solvent mixture (hexane/DCM 1/1) to afford **1,4-bis(3-((3-bromophenyl)ethynyl)phenyl)-8,8-dimethoxy-4,4a,5,7a-tetrahydro-1H-1,4-methanocyclopenta[d]pyridazine (6)** as a brown solid (2.1134 g, 3.0 mmol, 100 %). ¹H NMR (CDCl₃, 400 MHz): δ 8.15 (m, 1H), 8.06 (m, 1H), 8.01-7.99 (m, 1H), 7.92-7.90 (m, 1H), 7.22 (s, 2H), 7.59-7.55 (m, 2H), 7.49-7.45 (m, 5H), 7.23 (t, *J* = 8.0 Hz, 1H), 5.62-5.60, 5.54-5.52 (m, 2H), 4.19 (m, 1H), 3.77-3.70 (m, 1H), 3.03 (s, 3H), 2.74 (s, 3H), 2.43-2.36, 2.26-2.17 (m, 2H). ¹³C NMR (100 MHz, CDCl₃): δ 32.25, 42.33, 52.00, 52.31, 52.33, 57.15, 88.01, 80.04, 90.76, 90.78, 92.48, 94.23, 118.42, 122.23, 123.00, 123.04, 125.29, 126.34, 128.44, 128.62, 128.66, 129.19, 129.83, 130.18, 131.18, 131.25, 131.41, 131.48, 131.90, 134.39, 134.54, 136.46, 136.64. HRMS (ESI) Calcd for C₃₈H₂₉O₂N₂Br₂ [M+H]⁺ = 703.05903, Found 703.05981.

A side-arm flask was charged with **6** (4.2359 g, 10.7 mmol), tetrakis (triphenylphosphine) palladium (0.6777 g, 0.59 mmol), and copper iodide (0.3094 g, 1.6 mmol) under nitrogen atm. 1-Bromo-3-iodobenzene-5-octylbenzene (10.7813 g, 27.3 mmol), THF (90 mL) and triethylamine

(40 mL) were added and refluxed at 65 °C for 14 h under nitrogen atm. After filtration by celite and concentration, then water was added and the organic product was extracted with CHCl₃, washed by aqueous NH₄Cl and brine, dried over MgSO₄, filtered and concentrated. The residue was separated by column chromatography on silica gel with a solvent mixture (hexane/EtOAc 10/1) to afford **1,4-bis(3-((3-bromo-5-octylphenyl)ethynyl)phenyl)-8,8-dimethoxy-4,4a,5,7a-tetrahydro-1H-1,4-methanocyclopenta[d]pyridazine (8)** as a yellow solid (9.8702 g, 10.6 mmol, 99 %). ¹H NMR (CDCl₃, 400 MHz): δ 8.14 (s, 1H), 8.05 (s, 1H), 8.00 (d, *J* = 4.0 Hz, 1H), 7.91 (d, *J* = 4.0 Hz, 1H), 7.58-7.44 (m, 6H), 5.61-5.51 (m, 2H), 4.21-4.19 (m, 1H), 3.76-3.71 (m, 1H), 3.03 (s, 3H), 2.73 (s, 3H), 2.39-2.36, 2.26-2.21 (m, 2H). ¹³C NMR (100 MHz, CDCl₃): δ 14.11, 22.67, 29.19, 29.21, 29.41, 31.10, 31.87, 32.25, 35.58, 42.28, 51.98, 52.28, 57.12, 88.32, 88.35, 90.25, 90.27, 92.47, 94.21, 118.37, 122.00, 123.08, 123.12, 124.83, 126.34, 128.35, 128.61, 128.65, 129.10, 130.33, 131.12, 131.20, 131.37, 131.57, 131.63, 131.84, 134.54, 136.39, 136.57, 145.19. HRMS (ESI) Cald for C₅₄H₆₀O₂N₂Br₂Na [M+Na]⁺ = 951.28888, Found 951.28933.

(h) To a solution of **6** (1.8429 g, 2.6 mmol) in THF (80 mL) was slowly added a hexane solution of *n*-BuLi (1.61 M) (4.0 mL, 6.4 mmol) and a solution of I₂ (3.9655 g, 15.6 mmol) in THF (20 mL) to the reaction mixture at -78 °C and stirred for 5.5 h. The reaction mixture was quenched by saturated aqueous NH₄Cl, extracted with Hexane/EtOAc, washed with Na₂S₂O_{3aq}, dried over MgSO₄, filtered and concentrated. The residue was separated by column chromatography on silica gel with a solvent mixture (hexane/EtOAc 8/1) to afford **1,4-bis(3-((3-iodophenyl)ethynyl)phenyl)-8,8-dimethoxy-4,4a,5,7a-tetrahydro-1H-1,4-methanocyclopenta[d]pyridazine (7)** as a brown solid (1.867 g, 2.3 mmol, 90 %). ¹H NMR (CDCl₃, 400 MHz): δ 8.17-8.12 (m, 1H), 8.08-7.86 (m, 4H), 7.69-7.67 (d, *J* = 8.0 Hz, 1H), 7.60-7.44 (m, 6H), 7.37-7.35 (m, 2H), 7.12-7.08 (t, *J* = 8.0 Hz, 1H), 5.61-5.52 (m, 2H), 4.19 (m, 1H), 3.77-3.71 (m, 1H), 3.03 (s, 3H), 3.01 (s, 3H), 2.43-2.21 (m, 2H). ¹³C NMR (100 MHz, CDCl₃): δ 32.26, 42.33, 52.00, 52.31, 52.34, 57.11, 57.14, 87.85, 87.87, 90.80, 90.82, 92.48, 93.70, 94.22, 118.42, 123.03, 123.07, 125.36, 126.34, 126.42, 128.35, 128.39, 128.42, 128.55, 128.62, 128.66, 129.18, 129.89, 130.15, 130.20, 130.74, 131.08, 131.16, 131.22, 131.38, 131.67, 131.80, 131.87, 134.50, 134.54, 136.64, 136.93, 137.34, 140.24. HRMS (ESI) Cald for C₃₈H₂₈O₂N₂I₂Na [M+Na]⁺ = 821.01324, Found 821.01398.

To a solution of **8** (5.4265 g, 5.8 mmol) in THF (120 mL) was slowly added a hexane solution of *n*-BuLi (2.6 M) (6.8 mL, 17.7 mmol) and a solution of I₂ (8.850 g, 34.9 mmol) in THF (20 mL) to the reaction mixture at -78 °C and stirred for 4.5 h. The reaction mixture was quenched by saturated aqueous NH₄Cl, extracted with Hexane/EtOAc, washed with Na₂S₂O_{3aq}, dried over MgSO₄, filtered and concentrated. The residue was separated by column chromatography on silica gel with a solvent mixture (hexane/DCM 1/1) to afford **1,4-bis(3-((3-iodo-5-octylphenyl)ethynyl)phenyl)-8,8-dimethoxy-4,4a,5,7a-tetrahydro-1H-1,4-methanocyclopenta[d]pyridazine (9)** as a brown solid (6.8930 g, 6.7 mmol, 67 %). ¹H NMR (CDCl₃, 400 MHz): δ 8.13 (m, 1H), 8.04 (m, 1H), 8.00-7.98 (m, 1H), 7.92-7.90 (m, 1H), 7.73 (m, 2H), 7.57-7.44 (m, 8H), 7.40-7.34 (m, 2H), 5.60-5.53 (m, 2H), 4.19 (m, 1H), 3.76-3.72 (m, 1H), 3.03 (s, 3H), 2.73 (s, 3H), 2.43-2.21 (m, 2H). ¹³C NMR (100 MHz, CDCl₃): δ 13.45, 14.10, 22.66, 29.19, 29.20, 29.28, 29.40, 29.47, 30.48, 31.07, 31.31, 31.87, 34.40, 34.74, 35.47, 35.76, 42.99, 43.35, 52.15, 52.28, 53.05, 66.00, 66.02, 88.25, 88.28, 88.85, 90.14, 90.16, 122.00, 122.04, 122.97, 123.01, 123.66, 124.60, 124.88, 125.84, 126.14, 128.13, 128.33, 128.43, 128.47, 128.80, 128.93, 129.00, 129.09, 129.18, 129.53, 129.75, 130.32, 130.34, 131.24, 131.33, 131.44, 131.56, 131.59, 131.79, 131.85, 131.41, 139.57, 145.15, 145.22. HRMS (ESI) Calcd for C₅₄H₆₀O₂N₂I₂Na [M+Na]⁺ = 1045.26364, Found 1045.26379.

(i) A side-arm flask was charged with **7** (0.6826 g, 0.85 mmol), ((3-ethynylphenyl)ethynyl)trimethylsilane (0.2045 g, 1.03 mmol), tetrakis (triphenylphosphine) palladium (0.0987 g, 0.085 mmol), and copper iodide (0.0241 g, 0.093 mmol) under nitrogen atm. THF (60 mL) and triethylamine (10 mL) were added and refluxed at 60 °C for 18.5 h under nitrogen atm. The organic product was extracted with CHCl₃, washed by aqueous NH₄Cl, dried over MgSO₄, filtered and concentrated. The residue was separated by column chromatography on silica gel with a solvent mixture (hexane/DCM 1/2) to afford **4-(3-((3-iodophenyl)ethynyl)phenyl)-8,8-dimethoxy-1-(3-((3-((trimethylsilyl)ethynyl)phenyl)ethynyl)phenyl)ethynyl)phenyl)-4,4a,5,7a-tetrahydro-1H-1,4-methanocyclopenta[d]pyridazine (11)** as a mixture (0.6512 g). It is difficult to isolate only **11** by column chromatography, hence it was directly used in the next step.

A side-arm flask was charged with **9** (6.2609 g, 6.1 mmol), ((3-ethynylphenyl)ethynyl)trimethylsilane (1.4241 g, 7.2 mmol), tetrakis (triphenylphosphine)

palladium (0.3685 g, 0.32 mmol), and copper iodide (0.1747 g, 0.92 mmol) under nitrogen atm. THF (100 mL) and triethylamine (20 mL) were added and refluxed at 60 °C for 16.5 h under nitrogen atm. After filtration by celite and concentration, the residue was separated by column chromatography on silica gel with a solvent mixture (hexane/EtOAc 10/1) to afford **4-(3-((3-iodo-5-octylphenyl)ethynyl)phenyl)-8,8-dimethoxy-1-(3-((3-octyl-5-((3-((trimethylsilyl)ethynyl)phenyl)ethynyl)phenyl)ethynyl)phenyl)-4,4a,5,7a-tetrahydro-1H-1,4-methanocyclopenta[d]pyridazine (13)** as a mixture (6.0591 g). It is difficult to isolate only **13** by column chromatography, hence it was directly used in the next step.

(j) A solution of **11** as a mixture (0.6411 g, 0.74 mmol), K₂CO₃ (0.4095 g, 2.96 mmol) in THF (20 mL) and MeOH (30 mL) was stirred at room temperature for 5 h. After filtration by celite and concentration, the residue was separated by column chromatography on silica gel with a solvent mixture (hexane/DCM 1/1) and by GPC to afford **1-(3-((3-((3-ethynylphenyl)ethynyl)phenyl)ethynyl)phenyl)-4-(3-((3-iodophenyl)ethynyl)phenyl)-8,8-dimethoxy-4,4a,5,7a-tetrahydro-1H-1,4-methanocyclopenta[d]pyridazine (12)** (0.2897 g, 0.36 mmol, 43 % (2step)). ¹H NMR (CDCl₃, 400 MHz): δ 8.16 (m, 1H), 8.07 (m, 1H), 8.01-7.99 (m, 1H), 7.93-7.90 (m, 1H), 7.74 (s, 1H), 7.67 (m, 1H), 7.58-7.45 (m, 9H), 7.38-7.30 (m, 3H), 7.11-7.07 (t, *J* = 8.0 Hz, 1H), 5.60-5.54 (m, 2H), 4.20 (m, 1H), 3.74 (m, 1H), 3.10 (s, 1H), 3.04 (s, 3H), 2.74 (s, 3H), 2.39-2.27 (m, 2H). ¹³C NMR (100 MHz, CDCl₃): δ 13.88, 22.13, 29.16, 32.25, 39.70, 42.28, 42.31, 42.3, 51.98, 52.28, 57.09, 57.13, 57.15, 63.33, 82.73, 88.65, 88.67, 88.94, 89.24, 89.45, 89.48, 89.56, 90.17, 90.19, 92.47, 92.49, 92.50, 93.68, 94.19, 94.22, 94.24, 118.40, 122.57, 123.19, 123.24, 123.36, 123.43, 123.50, 123.54, 123.68, 125.34, 126.35, 126.40, 128.07, 128.32, 128.37, 128.41, 128.48, 128.54, 128.57, 128.62, 128.69, 128.83, 129.08, 129.16, 129.87, 130.21, 130.72, 131.06, 131.14, 131.20, 131.37, 131.53, 131.65, 131.77, 131.85, 131.88, 131.98, 134.47, 134.52, 134.70, 135.16, 136.34, 136.40, 136.44, 136.51, 136.58, 136.62, 137.71, 140.22. HRMS (ESI) Calcd for C₄₈H₃₃O₂N₂INa [M+Na]⁺ = 819.14789, Found 819.14807.

A solution of **13** as a mixture (6.7828 g, 6.2 mmol), K₂CO₃ (3.4392 g, 25 mmol) in THF (30 mL) and MeOH (40 mL) was stirred at room temperature for 4 h. After filtration by celite and concentration, the residue was separated by column chromatography on silica gel with a solvent mixture (hexane/DCM 1/1, hexane/EtOAc 20/1) to afford **1-(3-((3-((3-ethynylphenyl)ethynyl)-5-octylphenyl)ethynyl)phenyl)-4-(3-((3-iodo-5-octylphenyl)ethynyl)phenyl)-8,8-dimethoxy-**

4,4a,5,7a-tetrahydro-1H-1,4-methanocyclopenta[d]pyridazine (14) (2.1296 g, 2.1 mmol, 34 % (2step)). ¹H NMR (CDCl₃, 400 MHz): δ 8.16 (m, 1H), 8.07 (m, 1H), 8.01-7.99 (m, 1H), 7.93-7.90 (m, 1H), 7.74 (s, 1H), 7.67 (m, 1H), 7.58-7.45 (m, 9H), 7.38-7.30 (m, 3H), 7.11-7.07 (t, *J* = 8.0 Hz, 1H), 5.60-5.54 (m, 2H), 4.20 (m, 1H), 3.74 (m, 1H), 3.10 (s, 1H), 3.04 (s, 3H), 2.74 (s, 3H), 2.39-2.27 (m, 2H). ¹³C NMR (100 MHz, CDCl₃): δ 14.12, 22.68, 29.21, 29.23, 29.28, 29.40, 29.45, 29.47, 29.47, 31.14, 31.16, 31.34, 31.87, 31.89, 32.26, 35.36, 35.56, 35.77, 42.28, 51.99, 52.28, 57.11, 82.76, 88.49, 89.00, 89.57, 89.71, 90.32, 92.46, 92.47, 92.49, 93.75, 94.23, 118.37, 122.52, 122.93, 123.07, 123.13, 123.28, 123.32, 123.38, 123.53, 123.59, 124.91, 126.34, 126.37, 126.41, 128.27, 128.35, 128.48, 128.59, 128.64, 128.69, 128.79, 128.93, 129.04, 130.95, 131.02, 131.09, 131.18, 131.34, 131.64, 131.72, 131.80, 131.88, 132.06, 134.55, 135.15, 136.35, 136.40, 136.53, 136.58, 137.44, 137.54, 143.15, 143.48, 145.18. HRMS (ESI) Calcd for C₆₄H₆₅O₂N₂INa [M+Na]⁺ = 1043.39829, Found 1043.39856.

16:

71-76 °C

¹H NMR (400 MHz, C₆D₆) δ 8.55 (m, 1H), 8.42 (m, 1H), 8.12-8.15 (m, 1H), 7.96-7.99 (m, 1H), 7.78 (m, 1H), 7.72-7.74 (m, 1H), 7.60-7.64 (m, 2H), 7.15-7.37 (m, 9H), 6.77 (t, *J* = 8.0 Hz, 2H), 5.32-5.45 (m, 2H), 3.96-4.00 (m, 1H), 2.66 (s, 2H), 2.62-2.64 (s, 3H), 2.46-2.49 (s, 3H), 2.31-2.34 (t, *J* = 8.0 Hz, 4H), 2.05-2.08 (m, 1H), 1.39-1.44 (m, 4H), 1.20-1.31 (m, 20H), 0.90 (t, *J* = 8.0 Hz, 6H).

¹³C NMR (100 MHz, C₆D₆) δ 13.96, 22.69, 29.16, 29.23, 29.43, 30.96, 31.90, 32.04, 35.39, 42.16, 51.29, 51.82, 57.38, 78.11, 82.58, 88.92, 89.42, 89.93, 90.21, 92.17, 93.92, 118.42, 122.91, 123.52, 123.54, 123.61, 123.69, 123.78, 123.80, 123.82, 126.19, 126.32, 128.30, 128.42, 128.58, 128.66, 128.69, 129.25, 131.06, 131.21, 131.27, 131.58, 131.65, 131.74, 131.84, 132.09, 132.44, 134.05, 135.24, 137.25, 137.58, 143.45, 143.46.

IR [cm⁻¹] *v*_{max} : 3293, 300, 2924, 2853, 1599, 1587, 1477, 1443, 1466, 1458, 1183, 1108

HRMS (ESI) for C₇₄H₇₀N₂O₂Na [M+Na]⁺ = 1041.53295, Found 1041.53333

(k) A side-arm flask was charged with tetrakis (triphenylphosphine) palladium (0.978 g, 0.085 mmol) and copper iodide (0.161 g, 0.084 mmol) under nitrogen atm. THF (50 mL) and triethylamine (20 mL) were added and refluxed at 60 °C. A solution of **12** (0.272 g, 0.34 mmol) in THF (10ml) was dropped into the reaction mixture by syringe pump for 125 min and refluxed for 4h. After filtration by celite and concentration, water was added. The organic product was extracted with CHCl₃, washed with aqueous NH₄Cl, dried over MgSO₄, filtered and concentrated. The residue was separated by column chromatography on silica gel with a solvent mixture (hexane/DCM 1/1) and recrystallized to afford **AZ2g** (0.076 g, 0.11 mmol, 32 %). ¹H NMR (CDCl₃, 400 MHz): δ 8.39-8.36 (m, 2H), 8.02-7.99 (m, 3H), 7.67-7.65 (m, 1H), 7.60-7.56 (m, 3H), 7.49-7.45 (m, 8H), 7.37-7.33 (m, 3H), 5.61-5.57 (m, 2H), 4.35 (m, 1H), 3.86-3.82 (m, 1H), 3.03 (s, 3H), 2.78 (s, 3H), 2.41-2.17 (m, 2H). ¹³C NMR (100 MHz, C₆D₆): δ 31.86, 40.86, 51.16, 51.63, 52.92, 54.72, 89.53, 89.92, 89.97, 90.77, 90.80, 93.10, 94.65, 118.83, 123.65, 123.78, 123.94, 126.35, 126.51, 126.95, 127.00, 129.99, 130.26, 130.32, 134.23, 135.46, 136.08, 136.64, 136.68, 137.45, 137.53, 137.96. HRMS (ESI) Calcd for C₄₈H₃₂O₂N₂Na [M+Na]⁺ = 691.23560, Found 691.23584.

A side-arm flask was charged with tetrakis (triphenylphosphine) palladium (0.152 g, 0.13 mmol) and copper iodide (0.0247 g, 0.13 mmol) under nitrogen atm. THF (80 mL) and triethylamine (20 mL) were added and refluxed at 60 °C. A solution of **14** (0.5377 g, 0.53 mmol) in THF (10ml) was dropped into the reaction mixture by syringe pump for 125 min and refluxed for 4h. After filtration by celite and concentration, water was added. The organic product was extracted with CH₂Cl₂, washed with aqueous NH₄Cl, dried over MgSO₄, filtered and concentrated. The residue was separated by column chromatography on silica gel with a solvent mixture (hexane/EtOAc 10/1) and by GPC to afford **AZ2h** (0.0788 g, 0.088 mmol, 17 %). ¹H NMR (400 MHz, C₆D₆): δ 8.99-8.98 (m, 2H), 8.07 (d, *J* = 8.0 Hz, 3H), 7.94 (s, 1H), 7.67 (t, *J* = 10.0 Hz, 2H), 7.46 (d, *J* = 8.0 Hz, 1H), 7.39-7.37 (m, 5H), 7.32 (s, 2H), 7.21-7.16 (m, 1H), 6.90 (t, *J* = 8.0 Hz, 1H), 5.51, 5.39 (m, 2H), 4.14 (m, 1H), 3.54 (m, 1H), 2.59 (s, 3H), 2.33 (s, 3H), 2.28-2.14 (m, 2H). ¹³C NMR (100 MHz, C₆D₆): δ 13.96, 22.69, 29.13, 29.22, 29.41, 31.00, 31.02, 31.90, 34.07, 35.48, 35.51, 37.93, 48.79, 49.11, 52.23, 52.61, 53.75, 89.53, 89.74, 90.55, 91.01, 91.44, 91.53, 91.72, 91.94, 99.72, 123.12, 123.44, 123.49, 123.57, 123.99, 124.00, 124.02, 124.04, 129.12,

129.37, 130.22, 130.30, 130.54, 130.64, 130.80, 131.09, 131.69, 132.01, 132.69, 140.57, 143.33, 143.45. HRMS (ESI) Calcd for $C_{64}H_{64}O_2N_2Na$ $[M+Na]^+$ = 915.48600, Found 915.48572.

Photochemical Denitrogenation of **AZ2e,h, 16**.

A solution (30 mM) of **AZ2e,h,16** (12 μ mol) in C_6D_6 (0.4 ml) was added into the NMR tube. After vacuum degassing, a solution was exposed to the light (λ = 396 nm) for 48 h at room temperature to afford **CP2e,h, CP16**. The product **CP2h** cannot be refined because it was unstable under air condition.

trans-CP2h: 1H NMR (C_6D_6 , 400MHz): δ 5.66 (m, 2H), 3.58 (m, 1H), 3.31 (s, 3H), 2.97 (m, 1H), 2.82 (s, 3H), 2.40 (m, 2H). ^{13}C NMR (100 MHz, C_6D_6): δ 13.96, 22.69, 29.13, 29.22, 29.41, 31.00, 31.02, 31.90, 34.07, 35.48, 35.51, 37.93, 48.79, 49.11, 52.23, 52.61, 53.75, 89.53, 89.74, 90.55, 91.01, 91.44, 91.53, 91.72, 91.94, 99.72, 123.12, 123.44, 123.49, 123.57, 123.99, 124.00, 124.02, 124.04, 129.12, 129.37, 130.22, 130.30, 130.54, 130.64, 130.80, 131.09, 131.69, 132.01, 132.69, 140.57, 143.33, 143.45.

trans-17: 1H NMR (400 MHz, C_6D_6) δ 7.78 (m, 2H), 7.73-7.69 (m, 3H), 7.53-7.49 (m, 2H), 7.33-7.24 (m, 8H), 7.14-7.11 (m, 2H), 7.02 (dd, J = 8.0 Hz, 2H), 6.78 (t, J = 8.0 Hz, 2H), 5.62 (m, 2H), 3.58 (m, 1H), 3.40 (s, 3H), 3.03 (m, 1H), 2.83 (s, 3H), 2.69 (s, 2H), 2.41-2.35 (m, 2H), 2.30 (t, J = 8.0 Hz, 4H), 1.38 (m, 4H), 1.31-1.19 (m, 22H), 0.92 (t, J = 8.0 Hz, 6H); ^{13}C NMR (100 MHz, C_6D_6): δ 13.98, 22.71, 29.19, 29.25, 29.44, 30.95, 31.92, 34.03, 35.38, 37.97, 48.69, 49.21, 52.20, 52.87, 53.86, 78.10, 82.60, 88.82, 90.01, 90.41, 99.47, 122.89, 123.17, 123.42, 123.83, 129.57, 129.82, 129.95, 130.83, 131.50, 131.59, 131.83, 131.86, 132.43, 132.60, 132.91, 133.65, 135.25, 135.54, 143.35; HRMS (ESI) Calcd for $C_{74}H_{70}O_2Na$ $[M+Na]^+$ = 1013.39895, Found 1013.39872.

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1H NMR (400 MHz, C_6D_6): δ 8.45 (s, 1H), 8.37 (s, 1H), 8.26 (s, 1H), 7.98 (d, J = 6.0 Hz, 2H), 7.47-7.60 (m, 4H), 7.31-7.37 (m, 5H), 6.88-6.96 (m, 4H), 5.56-5.59 (m, 2H), 4.36-4.38 (m, 1H), 3.62-3.68 (m, 1H), 3.01-3.10 (m, 1H), 2.33-2.37 (m, 5H), 1.39-1.45 (m, 5H), 1.22-1.36 (m, 30H), 0.92 (t, J = 8.0 Hz, 9H); ^{13}C NMR (100 MHz, C_6D_6) δ 13.97, 22.70, 22.96, 23.82, 28.94, 29.14,

29.23, 29.43, 29.76, 29.82, 30.48, 30.96, 31.90, 35.46, 38.88, 51.82, 56.09, 67.70, 89.36, 89.86, 89.89, 89.98, 90.00, 90.75, 123.62, 13.65, 123.69, 123.73, 124.01, 124.04, 124.10, 130.30, 130.42, 130.83, 131.06, 131.49, 132.16, 132.94, 134.07, 134.58, 134.78, 135.10, 137.74, 138.67, 139.37, 143.53, 143.56, 196.75, 198.81

HRMS (ESI) for $C_{61}H_{59}O_2$ $[M+H]^+ = 823.45096$, Found 823.45117.

19:

1H NMR (400 MHz, C_6D_6); δ 8.51 (s, 1H), 8.30 (s, 1H), 8.03 (s, 1H), 7.92 (s, 1H), 7.81 (s, 1H), 7.57 (dd, $J = 8.0$ Hz, 2H), 7.26-7.37 (m, 8H), 7.09 (t, $J = 8.0$ Hz, 1H), 6.99 (t, $J = 8.0$ Hz, 1H), 6.89 (t, $J = 8.0$ Hz, 1H), 5.15 (m, 1H), 4.86 (m, 1H), 4.54 (m, 1H), 3.94 (m, 1H), 3.26 (s, 3H), 3.12 (s, 3H), 2.31-2.34 (m, 5H), 2.01-2.19 (m, 2H), 1.41 (m, 5H), 1.21-1.29 (m, 30H), 0.92 (t, $J = 8.0$ Hz, 9H); ^{13}C NMR (100 MHz, C_6D_6); δ 13.97, 22.70, 29.15, 29.23, 29.43, 30.96, 30.98, 31.90, 35.44, 35.46, 36.59, 47.13, 50.81, 51.72, 62.43, 89.51, 89.74, 90.22, 90.31, 90.42, 90.74, 90.94, 91.68, 91.96, 112.85, 123.46, 123.53, 123.91, 124.09, 124.12, 126.76, 128.20, 128.27, 128.34, 128.49, 128.54, 128.83, 128.98, 129.13, 129.48, 129.75, 131.44, 131.70, 132.24, 132.97, 134.48, 135.20, 139.24, 140.01, 140.75, 143.39, 143.46, 173.20; HRMS (ESI) for $C_{64}H_{64}O_4Na$ $[M+Na]^+ = 919.46968$, Found 919.46960.

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1H NMR (400 MHz, C_6D_6); δ 8.36 (s, 1H), 8.27 (s, 2H), 8.11 (s, 1H), 8.00-8.03 (m, 2H), 7.77 (d, $J = 4.0$ Hz, 1H), 7.67 (d, $J = 4.0$ Hz, 2H), 7.43 (s, 2H), 7.32-7.37 (m, 5H), 6.96 (s, 2H), 6.91 (t, $J = 8.0$ Hz, 1H), 5.69-5.71 (m, 1H), 5.50-5.51 (m, 1H), 3.52-3.57 (m, 1H), 3.49 (s, 3H), 3.19 (s, 3H), 2.96-3.02 (m, 2H), 2.33-2.40 (m, 5H), 1.98-2.03 (m, 1H), 1.40 (m, 5H), 1.20-1.35 (m, 30H), 0.92 (t, $J = 8.0$ Hz, 10H); ^{13}C NMR (100 MHz, C_6D_6) δ 13.96, 22.70, 29.19, 29.23, 29.43, 30.98, 31.91, 35.49, 49.58, 50.39, 52.77, 52.88, 53.58, 89.45, 89.74, 89.85, 90.41, 91.01, 91.82, 110.02, 123.53, 123.68, 123.90, 123.97, 124.01, 124.11, 132.46, 134.40, 134.64, 134.91, 137.88, 139.26, 143.56, 143.65; HRMS (ESI) for $C_{64}H_{64}O_4Na$ $[M+Na]^+ = 919.46968$, Found 919.46960.

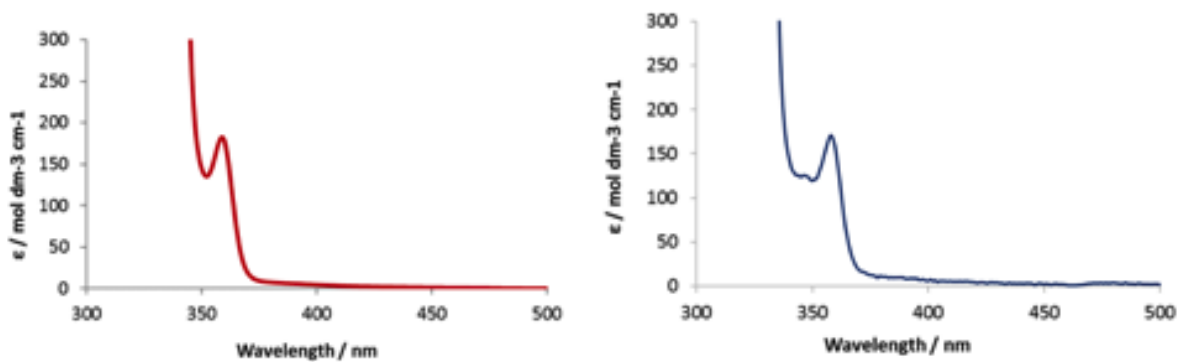


Figure S1. UV-vis spectrum of **AZ2g** (2.5 mM) (red line) and **AZ2h** (0.25 mM) (blue line) in benzene.

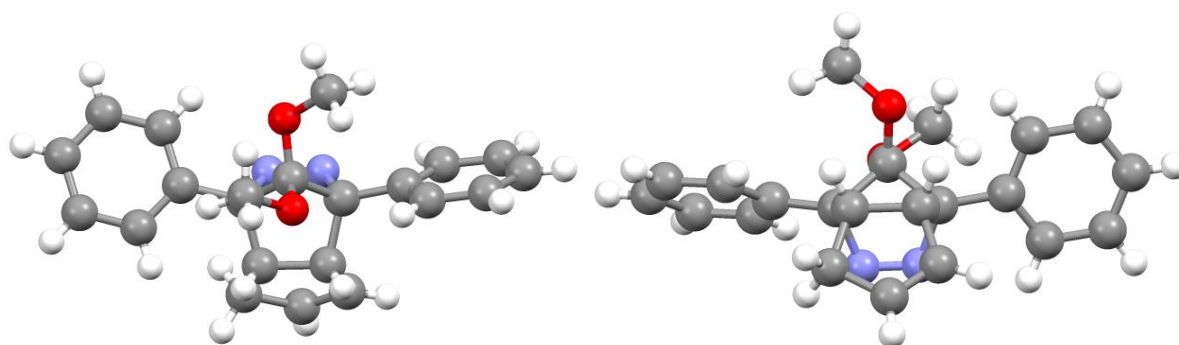


Figure S2 Top and side views of X-ray crystallographic structures of **AZ2e**.

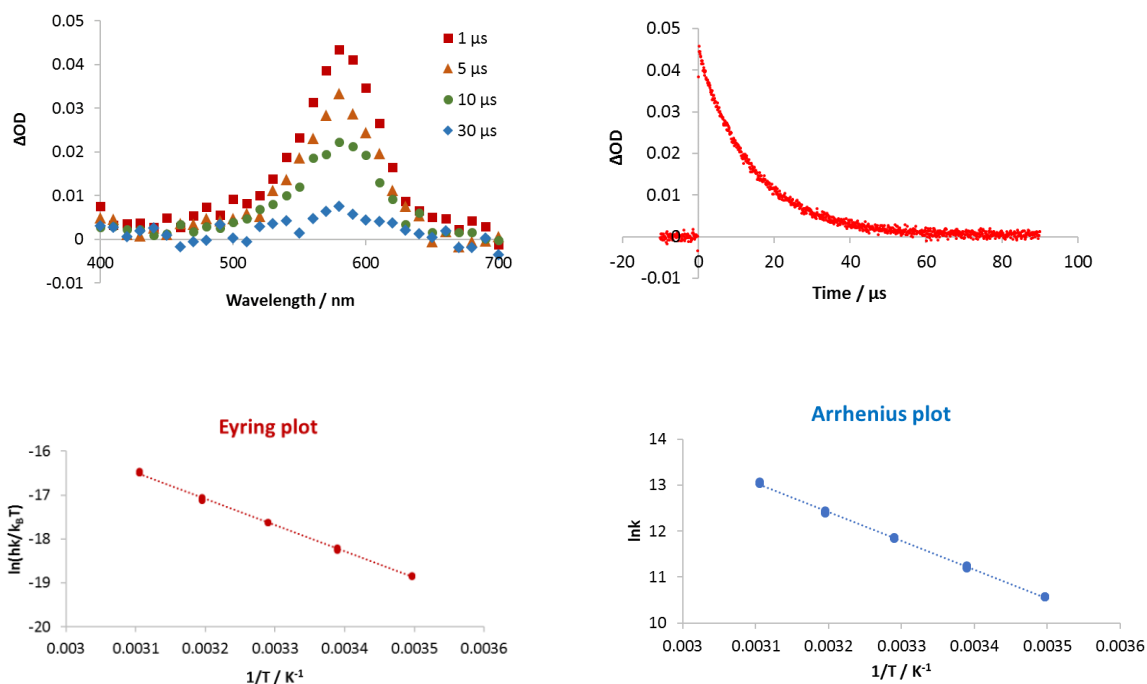


Figure S3. (a) Transient absorption spectra of **AZ2h** (8 mg, 3.0 mM) by laser flash photolysis (LFP) with YAG laser ($\lambda_{\text{exc}} = 355$ nm) in benzene under air at 293 K. (b) Absorbance change monitored at 580 nm. (c) Eyring plot. (d) Arrhenius plot.

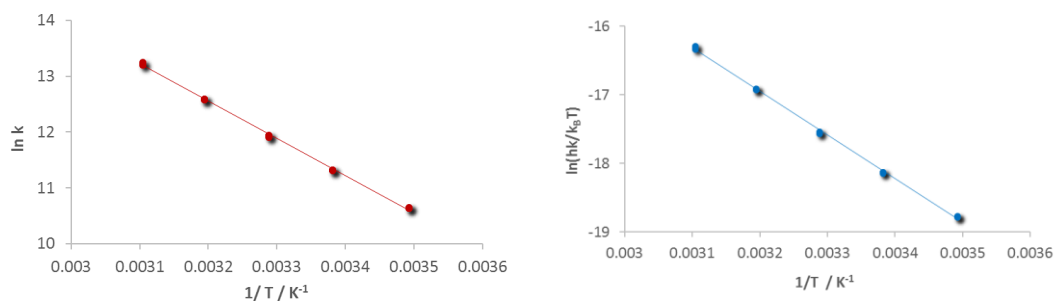


Figure S4. (a) Arrhenius plot for the decay of S-DR2g. (b) Eyring plot for the decay of S-DR2g.

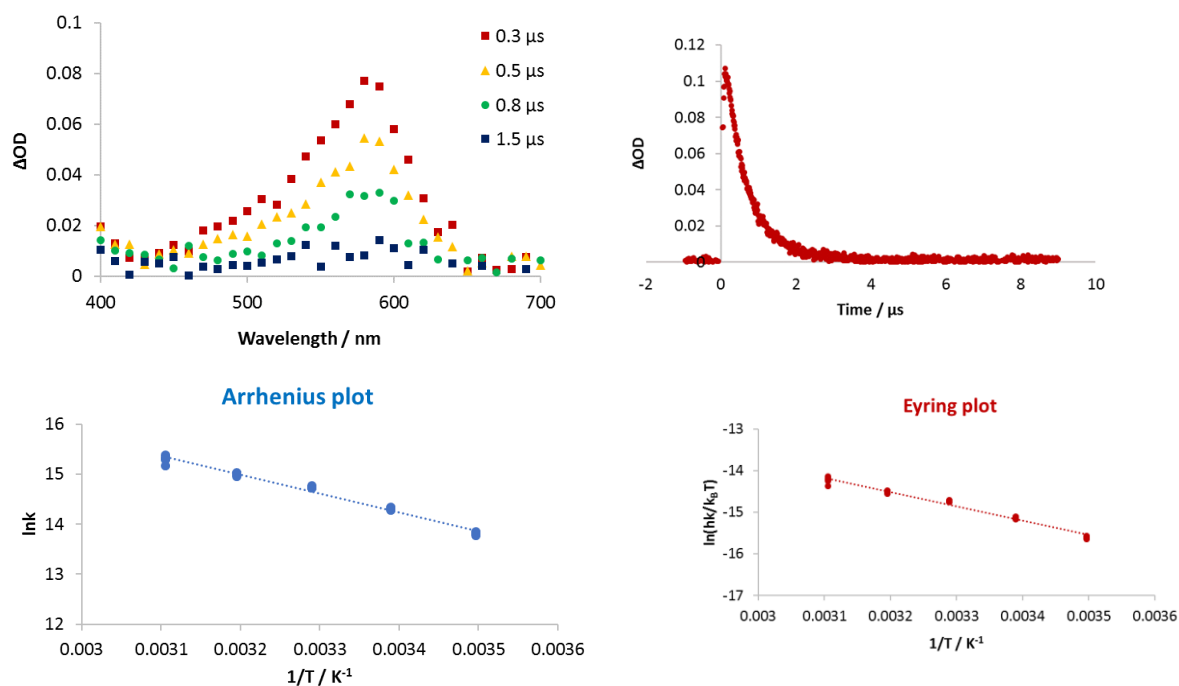


Figure S5. (a) Transient absorption spectra in the LFP of **16** (9.8 mg, 3.2 mM) with YAG laser ($\lambda_{exc} = 355$ nm) in benzene at 293 K. (b) Absorbance change monitored at 580 nm. (c) Arrhenius plot. (d) Eyring plot.

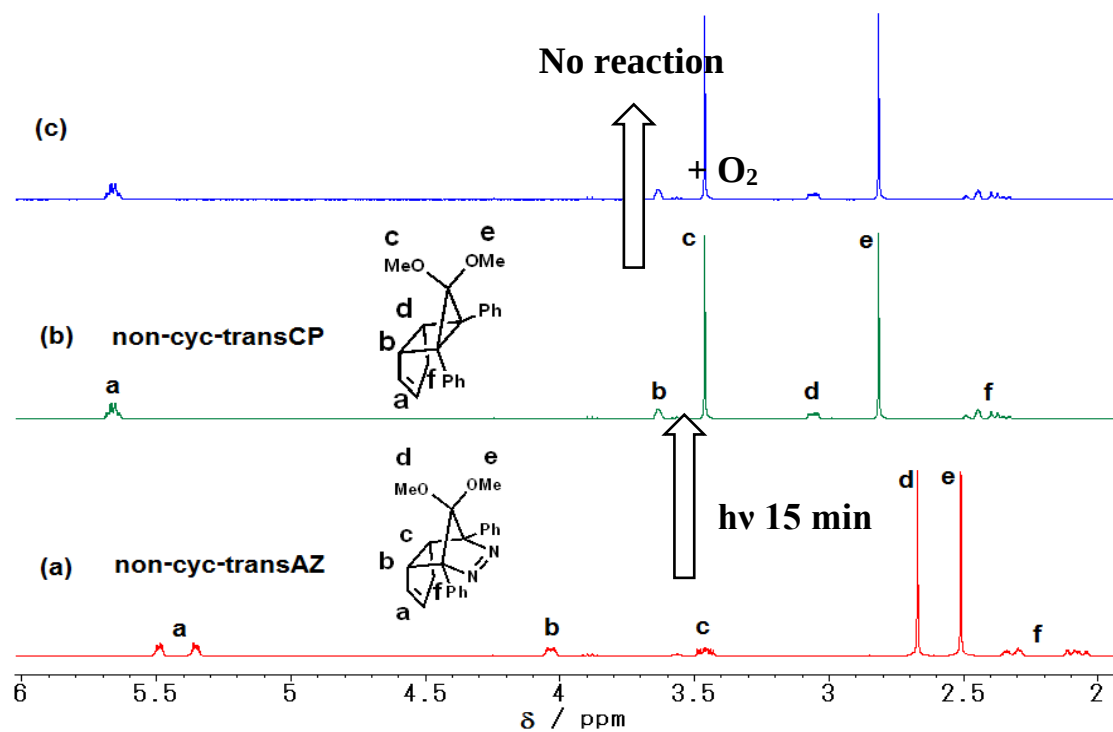


Figure S6. In-situ NMR (400 MHz) analysis of the photochemical denitrogenation of **AZ2e** (4.2 mg, 30 mM) in d_6 -benzene under degassing conditions with LED lamp (365 nm) at r.t. (a) ^1H NMR spectrum of **AZ2e** before irradiation. (b) ^1H NMR spectrum after 15 min of irradiation, and generated trans-**CP2e**. (c) ^1H NMR spectrum after 24 h under air condition at r.t.

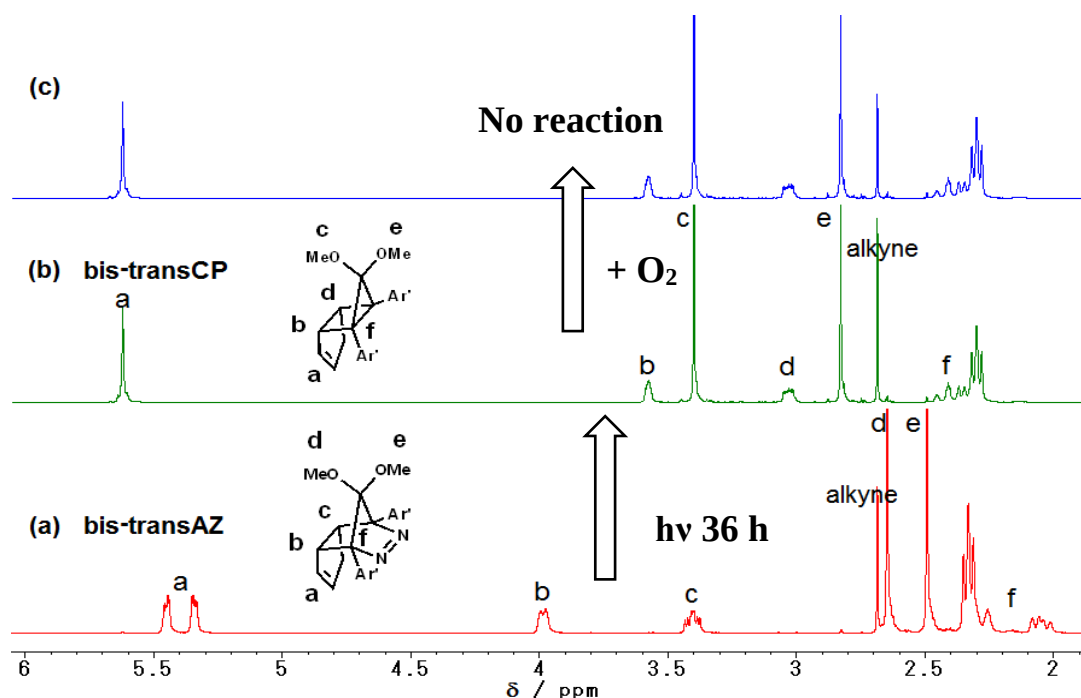


Figure S7. In-situ NMR (400 MHz) analysis of the photochemical denitrogenation of **16** (10 mg, 25 mM) in d_6 -benzene under degassing conditions with Xe lamp (398 ± 10 nm) at r.t. (a) ^1H NMR spectrum of **16** before irradiation. (b) ^1H NMR spectrum after 36 h of irradiation, and generated trans-**17**. (c) ^1H NMR spectrum after 25 h under air condition at r.t.

Computational details.

All structures were optimized in gas phase at the (U)B3LYP level of theory with the 6-31G(d) basis set using the Gaussian 09 program package.^[S1] Cartesian coordinates and energies for the optimized structures at the B3LYP/6-31G(d) level of theory were listed below. The frequency calculations confirmed the energy minimum structures and transition state structures.

AZ2e				
Multiplicity = 1			E = -1111.436555 Hartree	
Symmetry constraint = C ₁			S**2 = 0.0000	
C	0.066093	3.681836	-0.066440	
H	0.093087	4.694190	-0.461165	
C	1.143636	2.936028	0.189787	
H	2.173233	3.244943	0.035672	
C	-1.234763	2.973746	0.234869	
H	-1.871716	2.900427	-0.655459	
H	-1.821705	3.512585	0.991408	
C	-0.781440	1.578740	0.752052	
H	-1.148489	1.385468	1.762023	
C	0.778887	1.580046	0.734691	
H	1.212783	1.399499	1.722916	
C	1.134694	0.333279	-0.169351	
C	-0.006581	-0.662694	0.274874	
C	-1.134949	0.357589	-0.149335	
C	-2.572836	-0.089719	-0.222862	
C	-5.246927	-0.953063	-0.426330	
C	-2.965331	-0.980951	-1.234584	
C	-3.547562	0.371014	0.673771	
C	-4.872274	-0.059325	0.576607	
C	-4.287798	-1.409492	-1.333089	
H	-2.223533	-1.333175	-1.942097	
H	-3.280341	1.077444	1.453783	
H	-5.611087	0.312110	1.282060	
H	-4.571001	-2.099724	-2.123569	
H	-6.278354	-1.286470	-0.505296	
C	2.588995	-0.069332	-0.155680	
C	5.316184	-0.765217	-0.108061	
C	3.370335	0.021730	-1.317091	
C	3.196473	-0.506519	1.032870	
C	4.546790	-0.854666	1.053777	
C	4.722425	-0.324730	-1.291318	
H	2.909436	0.366664	-2.235794	

H	2.603897	-0.587508	1.938517
H	4.998841	-1.195143	1.981981
H	5.312253	-0.247445	-2.201161
H	6.369053	-1.034206	-0.089576
O	0.130678	-0.925925	1.644916
O	-0.167924	-1.826449	-0.475142
C	-0.891400	-1.725487	2.249639
H	-0.510978	-1.998153	3.236552
H	-1.089038	-2.628792	1.665484
H	-1.824572	-1.163532	2.359627
C	0.890402	-2.787852	-0.453628
H	1.697755	-2.498890	-1.133651
H	0.446830	-3.727288	-0.792754
H	1.297460	-2.916595	0.554157
N	0.633682	0.701492	-1.519172
N	-0.614278	0.723198	-1.519803

S-DR2e			
Multiplicity = 1		$E = -1001.918716$ Hartree	
Symmetry constraint = C_1		$S^{**2} = 0.2883$	
C	0.019551	-0.592320	0.121460
C	-1.179939	0.271583	-0.220590
C	-0.781493	1.624833	-0.742455
C	0.780506	1.648961	-0.691380
C	1.208920	0.280949	-0.225204
H	-1.127163	1.749994	-1.778361
H	1.210490	1.873964	-1.681196
C	1.093073	2.775945	0.281165
C	-1.269949	2.847869	0.112386
H	-1.777310	3.596229	-0.512931
H	-1.993892	2.548576	0.880705
C	-0.002929	3.402822	0.710284
C	2.569609	-0.150730	-0.154070
C	5.279666	-0.984518	-0.023898
C	3.628705	0.681307	-0.616730

C	2.928467	-1.424307	0.376600
C	4.256221	-1.822484	0.436858
C	4.953119	0.269212	-0.552070
H	3.400194	1.654457	-1.039822
H	2.146685	-2.078757	0.740240
H	6.316738	-1.304577	0.027391
C	-2.534301	-0.184284	-0.142934
C	-5.228183	-1.068859	-0.010045
C	-2.872387	-1.447605	0.423825
C	-3.605418	0.610989	-0.640194
C	-4.921982	0.174231	-0.574169
C	-4.192438	-1.871154	0.484875
H	-2.081073	-2.075116	0.813479
H	-3.391950	1.574536	-1.091851
H	-6.259042	-1.408315	0.042199
O	0.024439	-1.852732	-0.563067
O	0.025688	-1.079964	1.470741
C	0.018277	-1.790144	-1.978275
H	-0.882735	-1.294886	-2.366242
H	0.026366	-2.827142	-2.322788
H	0.906492	-1.278026	-2.373698
C	0.004444	-0.093657	2.488558
H	-0.911162	0.513064	2.454483
H	0.872518	0.576391	2.434539
H	0.032446	-0.640909	3.434017
H	-0.008537	4.228943	1.417444
H	2.109133	3.007073	0.586808
H	-4.421171	-2.838705	0.924592
H	-5.716422	0.804004	-0.966710
H	4.501164	-2.798027	0.849341
H	5.737985	0.926958	-0.917014

T-DR2e	
Multiplicity = 3	$E = -1001.914981$ Hartree
Symmetry constraint = C_1	$S^{**2} = 2.0017$

C	0.021093	-0.585713	0.117983
C	-1.190982	0.274800	-0.216678
C	-0.784535	1.623229	-0.751759
C	0.783246	1.650992	-0.688669
C	1.222124	0.282952	-0.226499
H	-1.128549	1.736732	-1.789555
H	1.222768	1.888771	-1.670662
C	1.082121	2.770493	0.297587
C	-1.278084	2.854907	0.087835
H	-1.765572	3.606511	-0.549853
H	-2.020511	2.564594	0.841505
C	-0.018615	3.400418	0.709176
C	2.576093	-0.150754	-0.151796
C	5.285895	-0.992668	-0.017705
C	3.641900	0.685490	-0.597991
C	2.931196	-1.433537	0.364624
C	4.257737	-1.834326	0.426495
C	4.963928	0.269237	-0.531438
H	3.417497	1.665032	-1.008575
H	2.146921	-2.092110	0.715018
H	6.322014	-1.315352	0.034988
C	-2.537678	-0.186155	-0.136129
C	-5.228926	-1.085530	0.001613
C	-2.869379	-1.455769	0.425619
C	-3.616917	0.606610	-0.626076
C	-4.929942	0.162238	-0.558013
C	-4.187058	-1.885052	0.489065
H	-2.074450	-2.082865	0.808260
H	-3.409952	1.573437	-1.073771
H	-6.257925	-1.430094	0.055776
O	0.022991	-1.833803	-0.580800
O	0.030462	-1.073499	1.461635
C	0.014917	-1.759879	-1.996812
H	-0.887181	-1.261822	-2.377488
H	0.022079	-2.794265	-2.348471

H	0.903464	-1.245305	-2.387127
C	0.010690	-0.091402	2.485321
H	-0.904531	0.515146	2.452851
H	0.880350	0.576093	2.432888
H	0.038544	-0.644238	3.427296
H	-0.031930	4.229347	1.412789
H	2.094048	2.997527	0.619599
H	-4.409663	-2.856104	0.924391
H	-5.728971	0.789725	-0.945091
H	4.498615	-2.815829	0.827300
H	5.752350	0.930014	-0.883275

trans-CP2e			
Multiplicity = 1		$E = -1001.932107$ Hartree	
Symmetry constraint = C_1		$S^{**2} = 0.0000$	
C	0.078564	-1.208943	-0.981438
C	0.823236	-0.554347	0.170134
C	0.867117	-1.541785	1.362851
C	-0.692260	-1.715508	1.306540
C	-0.754682	-0.679708	0.155307
H	1.484042	-2.436878	1.210303
H	-1.054072	-2.701766	1.009051
C	-1.248565	-1.270673	2.678924
H	-2.032041	-0.507884	2.581028
H	-1.708747	-2.112503	3.216437
C	1.095087	-0.899884	2.710512
C	-0.033272	-0.751373	3.408692
C	-1.845261	0.331307	0.036863
C	-3.982172	2.166494	-0.054364
C	-1.707702	1.620926	0.570728
C	-3.084153	-0.031049	-0.519358
C	-4.141231	0.877115	-0.568154
C	-2.764126	2.533223	0.518147
H	-0.766955	1.905621	1.031134
H	-3.218300	-1.037403	-0.908524

H	-4.804343	2.876119	-0.094351
C	1.793877	0.561306	-0.018341
C	3.735566	2.587808	-0.356219
C	1.491087	1.743920	-0.714441
C	3.095554	0.415693	0.491814
C	4.055613	1.413846	0.325695
C	2.449398	2.743816	-0.876142
H	0.504058	1.874193	-1.140531
H	3.365894	-0.496632	1.016232
H	4.479317	3.369917	-0.483882
O	0.116445	-2.586708	-1.197893
O	0.156526	-0.458825	-2.152251
C	1.347438	-3.042215	-1.760268
H	1.516043	-2.616027	-2.755608
H	1.268095	-4.129261	-1.834049
H	2.196758	-2.779272	-1.116744
C	-0.850478	-0.732290	-3.128701
H	-0.522241	-0.233543	-4.043697
H	-1.822798	-0.329905	-2.825849
H	-0.943099	-1.808326	-3.311035
H	2.077763	-0.592777	3.055049
H	-0.092465	-0.315108	4.402856
H	-2.633663	3.530443	0.930570
H	-5.090393	0.577483	-1.005211
H	2.187336	3.650503	-1.415920
H	5.054236	1.271197	0.731216

cis-CP2e			
Multiplicity = 1		$E = -1001.924399$ Hartree	
Symmetry constraint = C_1		$S^{**2} = 0.0000$	
C	0.164529	0.882599	0.907708
C	-0.752109	0.541209	-0.250255
C	-0.675550	1.577251	-1.397301
C	0.859738	1.364511	-1.502849
C	0.839683	0.342507	-0.320644

H	-1.210448	1.198343	-2.271963
H	1.176786	0.838490	-2.410708
O	0.243257	2.197815	1.341262
O	0.179047	-0.039592	1.956119
C	1.492204	2.537615	1.944595
H	1.666787	1.955274	2.856818
H	1.432261	3.599007	2.193454
H	2.321947	2.377892	1.245249
C	-0.820710	0.188222	2.956985
H	-0.637784	1.125370	3.492056
H	-0.745425	-0.653294	3.648193
H	-1.824593	0.213744	2.519804
C	-1.847178	-0.462173	-0.177323
C	-3.998793	-2.300576	-0.103614
C	-1.661266	-1.757939	0.335087
C	-3.117017	-0.101530	-0.659416
C	-4.164918	-1.015065	-0.612069
C	-2.732462	-2.645210	0.364757
H	-0.695033	-2.064703	0.710915
H	-3.289252	0.889571	-1.063099
H	-4.819911	-3.005358	-0.076020
C	1.660728	-0.900028	-0.350067
C	3.300233	-3.195114	-0.513820
C	2.602292	-1.173954	0.653386
C	1.542668	-1.786088	-1.432930
C	2.357954	-2.911477	-1.500089
C	3.401631	-2.308810	0.556335
H	2.698154	-0.513137	1.504931
H	0.813672	-1.602963	-2.214353
H	3.929078	-4.074220	-0.575392
C	-0.910366	3.097416	-1.194352
H	-1.561557	3.515431	-1.973988
H	-1.375360	3.328753	-0.229083
C	1.448131	2.746406	-1.403461
C	0.491229	3.672633	-1.273390

H	2.513239	2.951608	-1.468333
H	0.674651	4.742514	-1.205202
H	2.260324	-3.580212	-2.329645
H	4.119292	-2.508924	1.324330
H	-2.581770	-3.629039	0.757533
H	-5.127394	-0.724353	-0.978164

trans- TS2e			
Multiplicity = 1		<i>E</i> = -1001.894583 Hartree	
Symmetry constraint = <i>C_I</i>		S**2 = 0.0001	
C	-0.110996	-0.689542	0.904114
C	-1.065527	0.341437	0.397160
C	-0.727062	1.709617	0.949915
C	0.779455	1.516767	1.269440
C	1.008123	0.221384	0.502213
H	0.956299	1.317353	2.338195
C	-2.349939	0.002450	-0.217554
C	-4.930302	-0.640458	-1.212779
C	-2.942694	-1.263590	0.021968
C	-3.108647	0.934144	-0.963719
C	-4.373024	0.616703	-1.452676
C	-4.206959	-1.574877	-0.467102
H	-2.402466	-1.988201	0.618990
H	-2.707921	1.917490	-1.170065
H	-5.917613	-0.886131	-1.594650
C	2.209358	-0.138124	-0.247432
C	4.595186	-0.800939	-1.626311
C	2.769505	-1.435846	-0.167894
C	2.880567	0.809061	-1.056042
C	4.052556	0.482383	-1.731127
C	3.945265	-1.755896	-0.841348
H	2.276375	-2.187337	0.436597
H	2.461844	1.803063	-1.161303
H	5.511028	-1.054536	-2.153432
O	-0.295867	-0.967324	2.284470

O	-0.130312	-1.960756	0.263011
C	-0.137115	-2.063617	-1.160977
H	0.647630	-2.767365	-1.455478
H	0.049948	-1.104104	-1.648648
H	-1.109522	-2.443672	-1.492417
C	0.733948	-1.728444	2.902687
H	0.748567	-2.761946	2.537681
H	0.517910	-1.726912	3.974478
H	1.724243	-1.278112	2.740057
H	-1.340050	1.946074	1.834035
C	-0.750709	2.949656	-0.009513
H	-0.834410	2.648093	-1.061914
H	-1.598446	3.613498	0.198901
C	1.437440	2.808173	0.856343
C	0.589687	3.609063	0.207862
H	2.470469	3.043866	1.095020
H	0.827902	4.596167	-0.181159
H	-4.635653	-2.550972	-0.254408
H	-4.925473	1.357157	-2.025622
H	4.541831	1.231459	-2.348540
H	4.358236	-2.757330	-0.749533

cis-TS2e			
Multiplicity = 1		<i>E</i> = -1001.901648 Hartree	
Symmetry constraint = <i>C</i> ₁		S**2 = 0.0000	
C	-0.037914	0.258762	0.861054
C	-1.071051	0.398918	-0.196900
C	-0.819218	1.620640	-1.053735
C	0.720939	1.601061	-1.094352
C	0.998079	0.385574	-0.216821
H	-1.235785	1.496580	-2.056638
H	1.097253	1.450596	-2.112672
C	1.137681	2.947800	-0.542039
C	-1.243918	3.001690	-0.455381
H	-1.870716	3.570268	-1.155498

H	-1.819867	2.888365	0.471570
C	0.079171	3.689392	-0.200759
C	2.199202	-0.427066	-0.378803
C	4.614828	-1.881567	-0.699613
C	2.834184	-0.540477	-1.639021
C	2.820894	-1.077857	0.715508
C	4.007785	-1.785972	0.555126
C	4.017665	-1.254384	-1.795838
H	2.385256	-0.074572	-2.510114
H	2.364810	-1.008430	1.695439
H	5.540731	-2.436978	-0.821644
C	-2.295251	-0.385651	-0.291723
C	-4.757100	-1.768261	-0.449117
C	-2.899929	-0.936603	0.863624
C	-2.965743	-0.551233	-1.524790
C	-4.175472	-1.234058	-1.601894
C	-4.113799	-1.611765	0.781907
H	-2.410472	-0.803306	1.821792
H	-2.522665	-0.153903	-2.432962
H	-5.703737	-2.298490	-0.509359
O	-0.021495	-0.981501	1.573170
O	-0.148437	1.244703	1.871355
C	-0.050729	-2.232519	0.887438
H	-0.032466	-2.114100	-0.198739
H	-0.964025	-2.768779	1.166109
H	0.821425	-2.821430	1.189873
C	0.980404	1.391386	2.721141
H	1.900273	1.554711	2.142604
H	1.112155	0.519458	3.373118
H	0.790925	2.275348	3.335769
H	2.177067	3.243900	-0.428362
H	0.135981	4.680999	0.243547
H	-4.565531	-2.012419	1.685802
H	-4.665970	-1.352752	-2.564604
H	4.475010	-1.324590	-2.779455

H	4.465558	-2.262941	1.418141
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AZ2g			
Multiplicity = 1		$E = -2108.034429$ Hartree	
Symmetry constraint = C_1		$S^{**2} = 0.0000$	
C	-3.623541	-1.462744	0.238306
C	-5.035545	-1.387630	0.937745
H	-5.767282	-1.850054	0.269029
C	-5.273822	0.149847	1.103946
H	-5.510087	-1.139099	4.302002
C	-5.397339	-0.973711	3.233776
C	-3.972500	0.763139	0.506008
H	-4.983322	-2.980504	2.570707
C	-5.125604	-1.928625	2.341171
N	-2.681106	-0.934741	1.259273
N	-2.868494	0.287796	1.418677
C	-3.722511	-0.206178	-0.716138
O	-2.560548	0.204781	-1.363915
O	-4.788943	-0.410470	-1.600830
C	-2.042183	-0.621279	-2.412640
H	-1.381438	-1.394119	-2.010347
H	-1.470311	0.043482	-3.064936
H	-2.845793	-1.095123	-2.983782
C	-5.122399	0.677096	-2.469757
H	-5.823418	0.268785	-3.201031
H	-4.238436	1.066823	-2.982678
H	-5.601638	1.494241	-1.920706
C	-3.198612	-2.802181	-0.310905
C	-2.343177	-5.265364	-1.383738
C	-1.915920	-3.288889	-0.041400
C	-4.051838	-3.565097	-1.123489
C	-3.625373	-4.788835	-1.643641
C	-1.466725	-4.507270	-0.585322
H	-1.249585	-2.712713	0.589504
H	-5.043858	-3.197272	-1.364609

H	-4.298825	-5.370416	-2.267478
H	-2.006395	-6.208363	-1.802982
C	-3.880142	2.256016	0.315717
C	-3.592750	5.041505	-0.017988
C	-5.001101	3.098492	0.315723
C	-2.613686	2.828629	0.158504
C	-2.451737	4.216188	-0.004947
C	-4.854331	4.477466	0.142588
H	-5.995525	2.686471	0.459396
H	-1.736900	2.193389	0.172944
H	-5.733738	5.115765	0.145011
H	-3.476741	6.113624	-0.142833
C	-1.135343	4.749024	-0.132147
C	-0.115335	-4.906233	-0.359842
C	0.015055	5.135020	-0.214003
C	1.062677	-5.148495	-0.179414
C	2.469526	-5.300248	0.004471
C	5.256303	-5.508421	0.333357
C	3.266330	-4.146227	0.088118
C	3.086583	-6.563353	0.090343
C	4.467712	-6.654869	0.256288
C	4.659322	-4.235094	0.246622
H	2.798490	-3.170324	0.020849
H	2.477613	-7.459595	0.027110
H	4.935926	-7.632952	0.323841
H	6.331852	-5.585522	0.457282
C	1.389168	5.508677	-0.281656
C	4.117374	6.190453	-0.380489
C	1.786590	6.849302	-0.456412
C	2.376715	4.516173	-0.160384
C	3.742287	4.843184	-0.206734
C	3.140368	7.176239	-0.504073
H	1.028799	7.620680	-0.550378
H	2.079311	3.482395	-0.023403
H	3.437746	8.212622	-0.637987

H	5.170827	6.449254	-0.416671
C	5.425168	-3.033558	0.300192
C	6.005209	-1.964921	0.321282
C	4.713341	3.807203	-0.075258
C	5.495403	2.883706	0.044519
C	6.611465	-0.674468	0.326707
C	7.750660	1.897529	0.313381
C	5.798874	0.464063	0.192236
C	8.004581	-0.508343	0.456858
C	8.558567	0.770325	0.449277
C	6.354763	1.754324	0.183245
H	4.725673	0.344706	0.092702
H	8.637550	-1.383794	0.561633
H	9.633717	0.889853	0.550078
H	8.186234	2.891645	0.307688
H	-6.124231	0.469695	0.498106
C	-5.524556	0.400725	2.618366
H	-6.519401	0.831876	2.795430
H	-4.801500	1.104913	3.048587

S-DR2g			
Multiplicity = 1		$E = -1998.518383$ Hartree	
Symmetry constraint = C_1		$S^{**2} = 0.3328$	
H	-6.294176	1.015912	-1.062442
C	-5.793085	0.700594	-0.136351
C	-6.630681	1.197323	1.095141
C	-5.763718	-0.858551	-0.047630
H	-7.568864	1.672451	0.775142
H	-6.090965	1.949515	1.683909
H	-6.329858	-1.316622	-0.875082
C	-6.895975	-0.058785	1.884226
H	-7.404151	-0.043545	2.845391
C	-6.444632	-1.158210	1.279373
H	-6.522814	-2.167744	1.671615
C	-4.312263	-1.258800	-0.112596

C	-4.351621	1.129700	-0.153916
C	-3.395075	-0.050122	-0.130924
C	-3.866072	-2.616306	-0.185415
C	-3.023862	-5.323857	-0.325938
C	-2.486032	-2.956286	-0.157301
C	-4.801457	-3.684913	-0.290417
C	-4.380841	-5.006907	-0.359599
C	-2.064860	-4.290787	-0.224458
H	-1.748446	-2.168763	-0.075883
H	-5.863629	-3.467526	-0.328178
H	-5.117308	-5.801810	-0.442408
H	-2.692354	-6.355925	-0.377680
C	-3.935640	2.495688	-0.270556
C	-3.153489	5.215535	-0.493968
C	-4.890526	3.533302	-0.465965
C	-2.567179	2.872284	-0.198411
C	-2.175394	4.213293	-0.305827
C	-4.499290	4.861841	-0.575015
H	-5.943882	3.285705	-0.542991
H	-1.815138	2.108497	-0.049911
H	-5.250139	5.632723	-0.726955
H	-2.845127	6.252812	-0.576107
C	-0.675509	-4.613154	-0.188045
C	0.504231	-4.906104	-0.156997
C	-0.796914	4.572159	-0.223782
C	0.373542	4.894775	-0.156413
C	1.756520	5.232296	-0.079362
C	4.494153	5.869429	0.067630
C	2.180980	6.576690	-0.066417
C	2.721805	4.214213	-0.017232
C	4.091950	4.519428	0.055297
C	3.538396	6.881636	0.007086
H	1.439560	7.367963	-0.114069
H	2.405205	3.177072	-0.027362
H	3.855416	7.920827	0.016752

H	5.551114	6.109708	0.123747
C	1.896741	-5.208807	-0.117704
C	4.652272	-5.777271	-0.036915
C	2.836551	-4.167495	-0.050997
C	2.355890	-6.541560	-0.143960
C	3.721875	-6.812492	-0.103467
C	4.215199	-4.438392	-0.010223
H	2.493282	-3.139037	-0.030723
H	1.634275	-7.350699	-0.195595
H	4.065594	-7.842993	-0.124122
H	5.715942	-5.990981	-0.005686
C	5.131528	-3.348670	0.055724
C	5.850701	-2.369796	0.110277
C	5.035897	3.452959	0.110767
C	5.781824	2.493612	0.151313
C	6.587481	-1.150947	0.167100
C	7.958874	1.303686	0.273160
C	5.881302	0.062703	0.136940
C	7.993378	-1.121027	0.251681
C	8.662340	0.100729	0.303794
C	6.552660	1.295230	0.188516
H	4.799076	0.047828	0.071774
H	8.546357	-2.054686	0.275677
H	9.746756	0.115597	0.369001
H	8.485155	2.252088	0.313753
O	-2.448966	-0.019492	0.944138
O	-2.475620	-0.054743	-1.228673
C	-2.989461	0.031699	2.254517
H	-3.562086	0.953554	2.426619
H	-2.131022	0.015632	2.929991
H	-3.634833	-0.829576	2.470947
C	-3.048833	-0.082481	-2.524668
H	-3.652013	-0.985666	-2.691724
H	-2.207577	-0.086130	-3.221619
H	-3.668740	0.803171	-2.721925

T-DR2g			
Multiplicity = 3		$E = -1998.514773$ Hartree	
Symmetry constraint = C_1		$S^{**2} = 2.0023$	
H	-6.289632	1.007058	-1.089324
C	-5.801877	0.693433	-0.155568
C	-6.660014	1.191522	1.061496
C	-5.767684	-0.870938	-0.055411
H	-7.603600	1.646586	0.727544
H	-6.137847	1.960464	1.644067
H	-6.335439	-1.339359	-0.875198
C	-6.911526	-0.057982	1.864801
H	-7.429781	-0.037538	2.820374
C	-6.445095	-1.159835	1.276461
H	-6.514316	-2.165422	1.680319
C	-4.315836	-1.277322	-0.113847
C	-4.361528	1.134503	-0.152484
C	-3.407302	-0.054982	-0.128089
C	-3.863182	-2.627325	-0.179839
C	-3.006727	-5.333183	-0.310841
C	-2.479605	-2.961496	-0.156104
C	-4.793764	-3.703745	-0.275698
C	-4.366141	-5.022750	-0.340450
C	-2.052853	-4.294140	-0.218269
H	-1.744922	-2.170613	-0.082028
H	-5.857287	-3.491941	-0.309196
H	-5.098688	-5.822040	-0.416240
H	-2.670109	-6.363713	-0.358701
C	-3.943544	2.494517	-0.266509
C	-3.154591	5.214868	-0.487260
C	-4.896565	3.536887	-0.460793
C	-2.572549	2.869540	-0.194860
C	-2.178753	4.209727	-0.300457
C	-4.501749	4.863597	-0.568888
H	-5.950549	3.291927	-0.537961
H	-1.821260	2.104633	-0.049019

H	-5.250722	5.636489	-0.720217
H	-2.843972	6.251508	-0.568128
C	-0.661971	-4.610564	-0.185189
C	0.518414	-4.901256	-0.156156
C	-0.799617	4.566520	-0.218924
C	0.370773	4.889653	-0.152299
C	1.753206	5.229701	-0.076418
C	4.489647	5.873094	0.068045
C	2.174765	6.575052	-0.064173
C	2.720900	4.213871	-0.014855
C	4.090431	4.522210	0.056403
C	3.531538	6.883104	0.008074
H	1.431519	7.364631	-0.111415
H	2.406703	3.175992	-0.024430
H	3.846200	7.923019	0.017202
H	5.546115	6.115750	0.123187
C	1.911181	-5.202971	-0.118893
C	4.667189	-5.770512	-0.041163
C	2.850762	-4.161438	-0.052720
C	2.370944	-6.535534	-0.146224
C	3.737058	-6.806009	-0.107272
C	4.229560	-4.431859	-0.013398
H	2.507217	-3.133090	-0.031550
H	1.649575	-7.344917	-0.197427
H	4.081092	-7.836391	-0.128740
H	5.730967	-5.983835	-0.011057
C	5.145374	-3.341694	0.052360
C	5.863762	-2.362248	0.107051
C	5.036926	3.457971	0.111315
C	5.785474	2.500642	0.151378
C	6.598517	-1.142215	0.164498
C	7.965119	1.315055	0.271995
C	5.890014	0.070093	0.135243
C	8.004356	-1.109623	0.248884
C	8.670915	0.113429	0.301707

C	6.558912	1.303907	0.187565
H	4.807814	0.053134	0.070221
H	8.559174	-2.042207	0.272175
H	9.755310	0.130371	0.366757
H	8.489577	2.264439	0.313171
O	-2.469233	-0.024323	0.947573
O	-2.496973	-0.055610	-1.228878
C	-3.011420	0.031088	2.258463
H	-3.579389	0.956027	2.426580
H	-2.153274	0.011890	2.933925
H	-3.659772	-0.827838	2.473671
C	-3.072419	-0.084469	-2.525316
H	-3.672219	-0.989824	-2.690176
H	-2.231549	-0.085360	-3.222441
H	-3.693851	0.800293	-2.719738

trans-CP2g			
Multiplicity = 1		$E = -1998.528329$ Hartree	
Symmetry constraint = C_i		$S^{**2} = 0.0000$	
C	6.229618	-0.512642	0.231650
C	4.816471	0.007991	0.584682
C	5.615648	-1.854470	-0.306678
C	4.187091	-1.308876	-0.027306
C	3.012339	-1.500948	-0.922812
C	0.853952	-1.922253	-2.681092
C	1.975972	-2.376217	-0.576546
C	2.946842	-0.832063	-2.156531
C	1.880203	-1.050118	-3.029529
C	0.886611	-2.588467	-1.440117
H	1.991993	-2.876912	0.385191
H	3.733183	-0.131267	-2.422119
H	1.844201	-0.526309	-3.980856
H	0.013520	-2.084699	-3.348609
C	4.419610	1.442156	0.529318
C	3.782407	4.192886	0.516653

C	5.395367	2.417351	0.803682
C	3.112551	1.868540	0.249167
C	2.780945	3.237653	0.250869
C	5.079481	3.775455	0.792420
H	6.408437	2.107219	1.042029
H	2.337571	1.140179	0.044605
H	5.850742	4.510595	1.006462
H	3.526322	5.247618	0.512395
C	1.443841	3.678471	0.013763
C	0.320466	4.118102	-0.144461
C	-0.217781	-3.405074	-1.046530
C	-1.208118	-4.035545	-0.728545
C	-2.489398	-4.539343	-0.341812
C	-5.074367	-5.369054	0.416576
C	-3.532324	-3.610838	-0.211044
C	-2.756064	-5.897503	-0.087247
C	-4.040332	-6.296245	0.285789
C	-4.825686	-4.006451	0.169454
H	-3.333925	-2.562655	-0.403495
H	-1.958911	-6.627778	-0.184693
H	-4.237915	-7.346997	0.479074
H	-6.068863	-5.689435	0.710810
C	-0.988397	4.663651	-0.303174
C	-3.565276	5.768220	-0.563389
C	-2.127270	3.853631	-0.161287
C	-1.159122	6.035742	-0.583039
C	-2.437646	6.572834	-0.711421
C	-3.419683	4.394873	-0.285482
H	-2.008902	2.797565	0.055303
H	-0.282472	6.665865	-0.694644
H	-2.556704	7.631032	-0.927146
H	-4.561286	6.188376	-0.660559
C	-5.798719	-2.970317	0.290065
C	-6.447298	-1.942137	0.337901
C	-4.554764	3.546387	-0.126407

C	-5.493313	2.786759	0.019673
C	4.103222	-1.019888	1.445890
O	4.859924	-1.720428	2.385403
O	2.821593	-0.701749	1.880413
C	2.779664	0.036339	3.107100
H	3.150312	-0.565986	3.943407
H	1.729058	0.285085	3.269384
H	3.367279	0.957847	3.039148
C	4.338151	-3.013221	2.692599
H	3.351368	-2.946323	3.165325
H	5.044977	-3.473859	3.386509
H	4.256968	-3.634244	1.790067
H	5.848730	-2.749714	0.274261
H	6.926235	-0.573804	1.076225
C	-6.483910	1.773723	0.183985
C	-8.345753	-0.317250	0.506966
C	-6.063948	0.435019	0.196152
C	-7.856071	2.055611	0.335157
C	-8.767452	1.012496	0.495512
C	-6.979867	-0.619102	0.354232
H	-5.010164	0.208866	0.079061
H	-8.194465	3.086903	0.326928
H	-9.823515	1.239216	0.612659
H	-9.062856	-1.122690	0.630094
C	6.059576	-1.998611	-1.779724
H	5.217076	-2.195775	-2.454222
H	6.751093	-2.844759	-1.902691
C	6.843100	0.102477	-1.003395
H	7.308450	1.083558	-1.000644
C	6.746031	-0.686381	-2.077062
H	7.130711	-0.439411	-3.063538

cis-CP2g	
Multiplicity = 1	$E = -1998.520361$ Hartree
Symmetry constraint = C_1	$S^{**2} = 0.0000$

C	-6.214271	-0.038386	0.155568
C	-4.781542	0.346084	-0.286934
C	-5.683642	-1.349303	0.794911
C	-4.225042	-1.029690	0.338278
H	-6.571626	0.673048	0.904491
H	-5.708896	-1.353470	1.890811
C	-3.063209	-1.304212	1.228789
C	-0.972045	-1.926428	3.011445
C	-2.036832	-2.174716	0.843304
C	-3.021475	-0.744295	2.516516
C	-1.987701	-1.057166	3.398625
C	-0.984069	-2.490178	1.721215
H	-2.036483	-2.599540	-0.153739
H	-3.801248	-0.052382	2.823638
H	-1.969971	-0.613832	4.390524
H	-0.159762	-2.167530	3.689889
C	-4.291872	1.750145	-0.188538
C	-3.508515	4.460823	-0.047469
C	-5.233298	2.789202	-0.303563
C	-2.945057	2.093249	0.012139
C	-2.542017	3.441516	0.069646
C	-4.844715	4.126510	-0.231853
H	-6.280283	2.547698	-0.461684
H	-2.198268	1.315365	0.109792
H	-5.591239	4.910680	-0.326138
H	-3.195677	5.498973	0.001601
C	-1.168730	3.802912	0.217465
C	-0.016901	4.185874	0.300848
C	0.094589	-3.325938	1.296195
C	1.049825	-3.991286	0.944369
C	2.286009	-4.565283	0.511039
C	4.783873	-5.552931	-0.343292
C	3.360443	-3.694615	0.278801
C	2.476373	-5.945178	0.309068
C	3.717830	-6.422438	-0.112869

C	4.611687	-4.170064	-0.149301
H	3.221275	-2.630328	0.430779
H	1.653406	-6.630680	0.485272
H	3.856137	-7.489299	-0.265061
H	5.745301	-5.933647	-0.673378
C	1.323604	4.668804	0.373694
C	3.960647	5.652127	0.467287
C	2.412055	3.807601	0.158272
C	1.575740	6.030668	0.642242
C	2.883367	6.507650	0.687728
C	3.733573	4.287993	0.199499
H	2.231346	2.758716	-0.049978
H	0.738321	6.700351	0.810449
H	3.064981	7.558534	0.895431
H	4.979327	6.025232	0.500296
C	5.627509	-3.192753	-0.366589
C	6.331606	-2.208866	-0.493584
C	4.813351	3.385185	-0.029649
C	5.696851	2.574278	-0.232270
C	-4.130763	-0.731905	-1.129232
O	-4.876934	-1.366692	-2.115817
O	-2.815740	-0.478260	-1.526945
C	-2.701249	0.242670	-2.758546
H	-3.058402	-0.356255	-3.602731
H	-1.637075	0.453979	-2.881140
H	-3.258482	1.184902	-2.725100
C	-4.442917	-2.695204	-2.403089
H	-3.422724	-2.704588	-2.804512
H	-5.134656	-3.088957	-3.150909
H	-4.485455	-3.325312	-1.506118
C	6.609440	1.499498	-0.448078
C	8.305875	-0.712486	-0.862194
C	6.108824	0.190164	-0.386901
C	7.978534	1.690343	-0.719791
C	8.807593	0.587781	-0.924116

C	6.941636	-0.923354	-0.589262
H	5.056705	0.034086	-0.177137
H	8.378944	2.697966	-0.769819
H	9.861995	0.744184	-1.134316
H	8.959514	-1.564373	-1.020847
C	-7.339785	-0.438958	-0.835422
H	-7.096098	-0.190038	-1.874825
H	-8.289126	0.061156	-0.598413
C	-6.531931	-2.440347	0.198361
H	-6.430130	-3.490793	0.458042
C	-7.440530	-1.941324	-0.647717
H	-8.176614	-2.536877	-1.183362

trans- TS2g			
Multiplicity = 1		$E = -1998.492856$ Hartree	
Symmetry constraint = C_1		$S^{**2} = 0.0000$	
H	-5.631585	-1.105366	2.148122
C	-5.474544	-0.528752	1.223607
C	-6.790736	-0.577928	0.369475
C	-5.241891	0.988624	1.448222
H	-7.459520	-1.385674	0.692042
H	-6.581970	-0.745542	-0.694586
H	-4.948057	1.243215	2.475838
C	-7.397442	0.794642	0.530908
H	-8.399909	1.034693	0.185095
C	-6.534546	1.658605	1.066924
H	-6.722333	2.713727	1.242560
C	-4.014121	1.111267	0.554222
C	-4.151593	-0.872681	0.547662
C	-3.063350	0.068670	1.005534
C	-3.660770	2.293852	-0.240826
C	-2.979467	4.656641	-1.632357
C	-2.388599	2.876105	-0.093182
C	-4.583826	2.919502	-1.104275
C	-4.242581	4.087752	-1.783460

C	-2.033876	4.047497	-0.785252
H	-1.667085	2.414691	0.570497
H	-5.565176	2.479096	-1.242055
H	-4.965934	4.554971	-2.446304
H	-2.710778	5.560206	-2.170733
C	-3.897412	-2.170250	-0.113248
C	-3.406149	-4.772196	-1.181720
C	-4.934290	-2.960976	-0.664199
C	-2.608103	-2.747337	-0.105410
C	-2.349212	-4.023659	-0.632124
C	-4.687734	-4.230782	-1.183725
H	-5.950111	-2.594937	-0.682933
H	-1.784167	-2.203764	0.333698
H	-5.513257	-4.804592	-1.596481
H	-3.214500	-5.759092	-1.590830
C	-0.712341	4.566709	-0.640420
C	0.438566	4.935606	-0.506263
C	-1.013976	-4.526178	-0.595184
C	0.143780	-4.895186	-0.544753
C	1.524283	-5.246749	-0.475758
C	4.262358	-5.880102	-0.326462
C	1.958188	-6.581804	-0.598066
C	2.480289	-4.236266	-0.279364
C	3.850666	-4.538166	-0.205237
C	3.316321	-6.884670	-0.521096
H	1.225236	-7.367800	-0.749876
H	2.155420	-3.205864	-0.186031
H	3.641797	-7.917004	-0.614756
H	5.319279	-6.120309	-0.269498
C	1.822006	5.250551	-0.354288
C	4.569342	5.788181	-0.062859
C	2.738746	4.202262	-0.171846
C	2.299096	6.575453	-0.388220
C	3.661515	6.830679	-0.240119
C	4.113489	4.455695	-0.030067

H	2.379843	3.179363	-0.146356
H	1.597000	7.391239	-0.528380
H	4.020609	7.855775	-0.265375
H	5.629707	5.991994	0.047360
C	5.000665	3.350618	0.127276
C	5.679534	2.347997	0.240073
C	4.783834	-3.476290	-0.019105
C	5.519016	-2.519302	0.130824
C	6.379768	1.110233	0.342323
C	7.687998	-1.375249	0.510194
C	5.657366	-0.087075	0.205932
C	7.769092	1.047461	0.567156
C	8.406712	-0.189234	0.649337
C	6.297648	-1.334823	0.285400
H	4.587810	-0.046903	0.032108
H	8.333987	1.968062	0.674014
H	9.478437	-0.229259	0.823038
H	8.190033	-2.335470	0.573366
O	-1.821579	-0.008972	0.316539
O	-2.731533	0.163943	2.372216
C	-1.754280	0.001137	-1.111954
H	-2.741979	-0.013323	-1.578483
H	-1.197933	-0.882879	-1.439595
H	-1.223105	0.901267	-1.435516
C	-2.108131	-0.978170	2.948069
H	-2.700828	-1.888345	2.784340
H	-2.045527	-0.783536	4.021841
H	-1.098344	-1.129883	2.550041

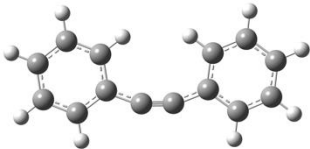
cis-TS2h	
Multiplicity = 1	$E = -1998.499547$ Hartree
Symmetry constraint = C_1	$S^{**2} = 0.0000$

H	-6.300182	1.249230	0.148491
C	-5.541195	0.850666	0.826425
C	-5.805289	1.327358	2.289405
C	-5.562610	-0.689741	0.841205
H	-6.698828	1.962778	2.353376
H	-4.967366	1.909382	2.692178
H	-6.317577	-1.098404	0.159753
C	-5.985310	0.028962	3.046077
H	-6.177851	0.010206	4.116718
C	-5.851172	-1.058234	2.279719
H	-5.927301	-2.085445	2.625953
C	-4.153263	-0.942773	0.303819
C	-4.146656	1.044403	0.255874
C	-3.180756	0.072626	0.827557
C	-3.822185	-2.156858	-0.450989
C	-3.236580	-4.589432	-1.791829
C	-2.566173	-2.780960	-0.318041
C	-4.776464	-2.790171	-1.279375
C	-4.485263	-3.986591	-1.930587
C	-2.258979	-3.980402	-0.982616
H	-1.811912	-2.330423	0.313421
H	-5.751218	-2.335540	-1.423894
H	-5.238478	-4.450489	-2.561748
H	-3.006207	-5.516365	-2.307403
C	-3.782334	2.230905	-0.523740
C	-3.114875	4.617982	-1.890098
C	-4.694793	2.846654	-1.407263
C	-2.529121	2.845738	-0.338854
C	-2.180686	4.024807	-1.019174
C	-4.362246	4.024003	-2.073617
H	-5.665585	2.392669	-1.581530
H	-1.819574	2.403144	0.349368
H	-5.079548	4.479527	-2.750971
H	-2.851619	5.529009	-2.418305
C	-0.948785	-4.528533	-0.839964

C	0.196280	-4.913415	-0.699821
C	-0.873098	4.566982	-0.835122
C	0.269666	4.945929	-0.664210
C	1.644351	5.273165	-0.465873
C	4.374955	5.835506	-0.080742
C	2.113783	6.600604	-0.505778
C	2.560322	4.234615	-0.229988
C	3.927006	4.500391	-0.041016
C	3.467920	6.868192	-0.311041
H	1.412194	7.408660	-0.687176
H	2.207152	3.209846	-0.199378
H	3.821153	7.895186	-0.341339
H	5.429053	6.048787	0.065704
C	1.571482	-5.255373	-0.533541
C	4.302322	-5.850291	-0.204764
C	2.500212	-4.230749	-0.286658
C	2.028348	-6.585454	-0.613853
C	3.382869	-6.869169	-0.447174
C	3.867140	-4.512737	-0.124676
H	2.157027	-3.203892	-0.225940
H	1.316899	-7.382717	-0.804315
H	3.726640	-7.897963	-0.508813
H	5.356535	-6.076186	-0.079699
C	4.770525	-3.433034	0.101967
C	5.472369	-2.455424	0.276856
C	4.817370	3.406708	0.169844
C	5.506854	2.417338	0.326875
C	6.212291	-1.249243	0.452856
C	7.608502	1.173365	0.767483
C	5.544715	-0.019546	0.326666
C	7.591310	-1.250980	0.741581
C	8.272733	-0.045110	0.896576
C	6.229705	1.197677	0.478716
H	4.483686	-0.009497	0.103247
H	8.113875	-2.197098	0.840717

H	9.335993	-0.055027	1.119599
H	8.144522	2.109597	0.886600
O	-2.949601	0.228243	2.206350
O	-1.880394	0.035223	0.239815
C	-2.339641	-0.877850	2.859612
H	-2.903304	-1.805286	2.688648
H	-2.355360	-0.649369	3.928141
H	-1.302591	-1.016197	2.533254
C	-1.679006	0.012916	-1.174241
H	-2.616046	-0.023957	-1.734511
H	-1.082658	-0.868763	-1.429725
H	-1.130865	0.914350	-1.466555

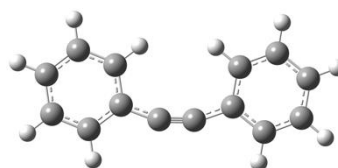
Cartesian coordinates and destabilization energy by angle-dependent energy change of 1,2-diphenylethyne at the B3LYP/6-31G(d) level of theory

$\theta = 159^\circ$ $E = -539.448416$ Hartree							
H	1.166123	3.562477	0.000000				
C	1.878981	2.743378	0.000000				
C	3.697478	0.628475	0.000000				
C	1.399100	1.419092	0.000000				
C	3.247176	3.001910	0.000000				
C	4.161854	1.946501	0.000000				
C	2.331416	0.361524	0.000000				
H	3.600883	4.029434	0.000000				
H	5.229424	2.148468	0.000000				
H	1.971458	-0.662358	0.000000				
H	4.403499	-0.197777	0.000000				
C	0.000000	1.116356	0.000000				
C	-1.023040	0.449499	0.000000				

C	-1.862812	-0.709803	0.000000
C	-3.454696	-3.028497	0.000000
C	-3.268311	-0.617661	0.000000
C	-1.268392	-1.988224	0.000000
C	-2.061101	-3.132245	0.000000
C	-4.054237	-1.767090	0.000000
H	-3.732362	0.364053	0.000000
H	-0.186123	-2.069254	0.000000
H	-1.587799	-4.110573	0.000000
H	-5.137325	-1.678587	0.000000
H	-4.068262	-3.925181	0.000000

$\theta = 160^\circ$

$E = -539.449750$ Hartree

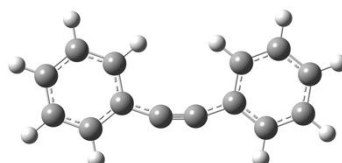


H	1.069886	3.570671	0.000000
C	1.813817	2.779713	0.000000
C	3.712933	0.736063	0.000000
C	1.385803	1.437793	0.000000
C	3.171112	3.090512	0.000000
C	4.125840	2.071148	0.000000
C	2.358173	0.416854	0.000000
H	3.485106	4.130870	0.000000
H	5.184782	2.314374	0.000000
H	2.037257	-0.620007	0.000000
H	4.450408	-0.062216	0.000000
C	0.000000	1.081603	0.000000
C	-1.008355	0.393516	0.000000
C	-1.843730	-0.768157	0.000000
C	-3.428021	-3.091757	0.000000
C	-3.249515	-0.680205	0.000000
C	-1.245132	-2.044658	0.000000
C	-2.034105	-3.191245	0.000000
C	-4.031601	-1.832237	0.000000

H	-3.716645	0.300038	0.000000
H	-0.162495	-2.121689	0.000000
H	-1.557792	-4.168104	0.000000
H	-5.114974	-1.747277	0.000000
H	-4.038843	-3.990317	0.000000

$\theta = 161^\circ$

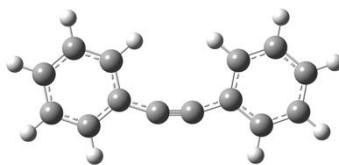
$E = -539.450966$ Hartree



H	-3.699684	0.232881	0.000000
C	-3.228152	-0.745243	0.000000
C	-2.001276	-3.251039	0.000000
C	-1.821989	-0.827071	0.000000
C	-4.004883	-1.900899	0.000000
C	-3.395630	-3.157705	0.000000
C	-1.217565	-2.100873	0.000000
H	-5.088632	-1.820874	0.000000
H	-4.002463	-4.058967	0.000000
H	-0.134529	-2.172637	0.000000
H	-1.520650	-4.225781	0.000000
C	-0.992688	0.338259	0.000000
C	0.000000	1.048087	0.000000
C	1.370371	1.457713	0.000000
C	4.083340	2.197323	0.000000
C	1.745378	2.815447	0.000000
C	2.381780	0.475405	0.000000
C	3.723052	0.847067	0.000000
C	3.089560	3.178723	0.000000
H	0.971063	3.576677	0.000000
H	2.101118	-0.573105	0.000000
H	4.491205	0.078280	0.000000
H	3.362838	4.230511	0.000000
H	5.131948	2.481854	0.000000

$\theta = 162^\circ$

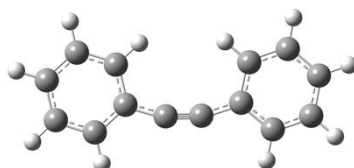
$E = -539.452162$ Hartree



H	-0.860890	-3.580408	0.000000
C	-1.667274	-2.853232	0.000000
C	-3.727834	-0.971529	0.000000
C	-1.351145	-1.480518	0.000000
C	-2.994606	-3.273863	0.000000
C	-4.029670	-2.336053	0.000000
C	-2.403830	-0.542546	0.000000
H	-3.222343	-4.336430	0.000000
H	-5.065075	-2.665414	0.000000
H	-2.168217	0.517019	0.000000
H	-4.528436	-0.236587	0.000000
C	0.000000	-1.013075	0.000000
C	0.974404	-0.278984	0.000000
C	1.795220	0.891652	0.000000
C	3.353919	3.232139	0.000000
C	3.201907	0.818538	0.000000
C	1.182672	2.161611	0.000000
C	1.959010	3.316750	0.000000
C	3.971137	1.979190	0.000000
H	3.679608	-0.156574	0.000000
H	0.099156	2.226147	0.000000
H	1.472296	4.288468	0.000000
H	5.055377	1.906096	0.000000
H	3.955069	4.137202	0.000000

$\theta = 163^\circ$

$E = -539.453234$ Hartree

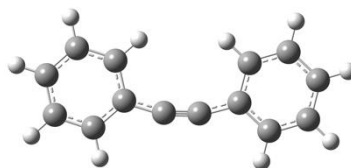


H	-3.657613	0.077045	0.000000
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C	-3.172354	-0.894324	0.000000
C	-1.909778	-3.382912	0.000000
C	-1.765102	-0.956583	0.000000
C	-3.932341	-2.061023	0.000000
C	-3.305291	-3.309115	0.000000
C	-1.142559	-2.221685	0.000000
H	-5.017123	-1.996482	0.000000
H	-3.899414	-4.218815	0.000000
H	-0.058530	-2.277430	0.000000
H	-1.415542	-4.350818	0.000000
C	-0.954815	0.220737	0.000000
C	0.000000	0.979561	0.000000
C	1.329410	1.504222	0.000000
C	3.967951	2.476136	0.000000
C	1.585171	2.889509	0.000000
C	2.422110	0.613171	0.000000
C	3.726077	1.099739	0.000000
C	2.892819	3.367764	0.000000
H	0.747733	3.580707	0.000000
H	2.232872	-0.455675	0.000000
H	4.558100	0.400556	0.000000
H	3.073769	4.439287	0.000000
H	4.987959	2.850445	0.000000

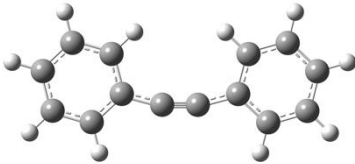
$\theta = 164^\circ$

$E = -539.454294$ Hartree



H	-3.630970	-0.014290	0.000000
C	-3.135695	-0.980586	0.000000
C	-1.847230	-3.456040	0.000000
C	-1.727822	-1.028386	0.000000
C	-3.883446	-2.155127	0.000000
C	-3.243442	-3.396649	0.000000
C	-1.092103	-2.286903	0.000000

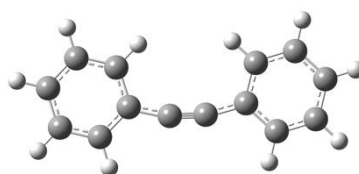
H	-4.968843	-2.101862	0.000000
H	-3.828131	-4.312437	0.000000
H	-0.007533	-2.331189	0.000000
H	-1.343089	-4.418818	0.000000
C	-0.931711	0.157979	0.000000
C	0.000000	0.944450	0.000000
C	1.302555	1.531458	0.000000
C	3.890161	2.631540	0.000000
C	1.489907	2.927640	0.000000
C	2.437609	0.695010	0.000000
C	3.716191	1.244935	0.000000
C	2.772582	3.469356	0.000000
H	0.619587	3.576928	0.000000
H	2.300709	-0.381808	0.000000
H	4.581571	0.587487	0.000000
H	2.900802	4.548465	0.000000
H	4.890563	3.055467	0.000000

$\theta = 165^\circ$ $E = -539.455256$ Hartree			
			
H	-3.602250	-0.108363	0.000000
C	-3.095234	-1.068542	0.000000
C	-1.776474	-3.528188	0.000000
C	-1.686842	-1.099283	0.000000
C	-3.828490	-2.252172	0.000000
C	-3.173318	-3.485761	0.000000
C	-1.035710	-2.349890	0.000000
H	-4.914457	-2.212180	0.000000
H	-3.746814	-4.408604	0.000000
H	0.049334	-2.380703	0.000000
H	-1.260732	-4.484794	0.000000
C	-0.907113	0.097401	0.000000
C	0.000000	0.911647	0.000000

C	1.272754	1.559832	0.000000
C	3.802997	2.785403	0.000000
C	1.391067	2.963570	0.000000
C	2.447406	0.779987	0.000000
C	3.697525	1.391921	0.000000
C	2.645659	3.567436	0.000000
H	0.489878	3.569252	0.000000
H	2.363209	-0.302241	0.000000
H	4.594216	0.777878	0.000000
H	2.720925	4.651529	0.000000
H	4.781327	3.258067	0.000000

$\theta = 166^\circ$

$E = -539.456184$ Hartree

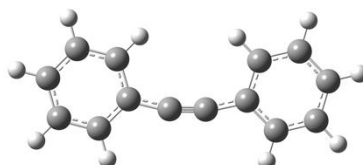


H	-0.351642	3.557445	0.000000
C	-1.283004	2.999290	0.000000
C	-3.667932	1.549075	0.000000
C	-1.237738	1.591250	0.000000
C	-2.504667	3.667234	0.000000
C	-3.700955	2.946161	0.000000
C	-2.451185	0.873255	0.000000
H	-2.523674	4.753767	0.000000
H	-4.653445	3.468964	0.000000
H	-2.422920	-0.211879	0.000000
H	-4.595286	0.982420	0.000000
C	0.000000	0.879325	0.000000
C	0.878889	0.035132	0.000000
C	1.639337	-1.173391	0.000000
C	3.090521	-3.581295	0.000000
C	3.048080	-1.163132	0.000000
C	0.969868	-2.414274	0.000000
C	1.693216	-3.603361	0.000000
C	3.763762	-2.357447	0.000000

H	3.569077	-0.210478	0.000000
H	-0.115527	-2.428950	0.000000
H	1.163583	-4.552342	0.000000
H	4.850200	-2.333462	0.000000
H	3.650483	-4.512421	0.000000

$\theta = 167^\circ$

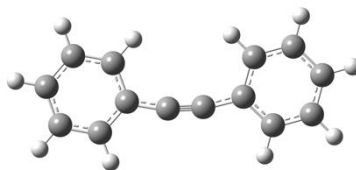
$E = -539.457016$ Hartree



H	-3.532849	-0.317274	0.000000
C	-2.995734	-1.260940	0.000000
C	-1.599080	-3.677767	0.000000
C	-1.586953	-1.247262	0.000000
C	-3.690769	-2.467404	0.000000
C	-2.996569	-3.679511	0.000000
C	-0.896245	-2.476427	0.000000
H	-4.777449	-2.462122	0.000000
H	-3.540505	-4.620103	0.000000
H	0.189256	-2.472286	0.000000
H	-1.053422	-4.617632	0.000000
C	-0.848556	-0.025620	0.000000
C	0.000000	0.848668	0.000000
C	1.198501	1.624017	0.000000
C	3.586587	3.107252	0.000000
C	1.169156	3.032570	0.000000
C	2.448190	0.971172	0.000000
C	3.627509	1.710368	0.000000
C	2.353823	3.764101	0.000000
H	0.209529	3.540620	0.000000
H	2.477247	-0.113947	0.000000
H	4.583590	1.193634	0.000000
H	2.315346	4.850116	0.000000
H	4.510103	3.679687	0.000000

$\theta = 168^\circ$

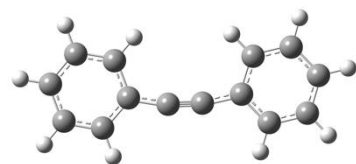
$E = -539.457809$ Hartree



H	-3.487766	-0.438587	0.000000
C	-2.931177	-1.370864	0.000000
C	-1.484461	-3.758435	0.000000
C	-1.522949	-1.328110	0.000000
C	-3.600790	-2.591576	0.000000
C	-2.881587	-3.789054	0.000000
C	-0.806849	-2.542679	0.000000
H	-4.687355	-2.608951	0.000000
H	-3.406017	-4.740652	0.000000
H	0.278342	-2.515740	0.000000
H	-0.919431	-4.686769	0.000000
C	-0.812180	-0.090609	0.000000
C	0.000000	0.817201	0.000000
C	1.151013	1.660841	0.000000
C	3.446428	3.283723	0.000000
C	1.037543	3.065171	0.000000
C	2.437467	1.083753	0.000000
C	3.570698	1.891857	0.000000
C	2.176580	3.865837	0.000000
H	0.049295	3.514940	0.000000
H	2.531081	0.002263	0.000000
H	4.555892	1.433112	0.000000
H	2.073502	4.947645	0.000000
H	4.334037	3.910402	0.000000

$\theta = 169^\circ$

$E = -539.458504$ Hartree

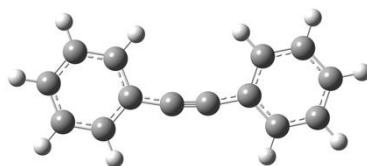


H	-3.437483	-0.565758	0.000000
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C	-2.858877	-1.484525	0.000000
C	-1.355313	-3.836876	0.000000
C	-1.451976	-1.408179	0.000000
C	-3.498979	-2.720902	0.000000
C	-2.751265	-3.900852	0.000000
C	-0.706996	-2.605180	0.000000
H	-4.584819	-2.764342	0.000000
H	-3.252839	-4.864694	0.000000
H	0.377250	-2.552312	0.000000
H	-0.768276	-4.751448	0.000000
C	-0.773457	-0.153031	0.000000
C	0.000000	0.787671	0.000000
C	1.098260	1.698394	0.000000
C	3.288732	3.459873	0.000000
C	0.898032	3.093069	0.000000
C	2.417813	1.201719	0.000000
C	3.498962	2.078310	0.000000
C	1.985420	3.962529	0.000000
H	-0.116164	3.480785	0.000000
H	2.577756	0.128028	0.000000
H	4.510708	1.681565	0.000000
H	1.815771	5.035907	0.000000
H	4.135965	4.140156	0.000000

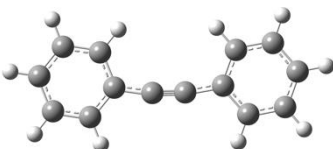
$\theta = 170^\circ$

$E = -539.459159$ Hartree



H	-3.380364	-0.697169	0.000000
C	-2.777903	-1.600465	0.000000
C	-1.213152	-3.912761	0.000000
C	-1.373438	-1.487372	0.000000
C	-3.385281	-2.853237	0.000000
C	-2.606931	-4.013236	0.000000
C	-0.597369	-2.664453	0.000000

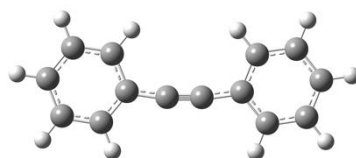
H	-4.469613	-2.925106	0.000000
H	-3.083122	-4.989868	0.000000
H	0.485133	-2.583178	0.000000
H	-0.602415	-4.811681	0.000000
C	-0.729896	-0.214292	0.000000
C	0.000000	0.760314	0.000000
C	1.039549	1.737049	0.000000
C	3.114201	3.633866	0.000000
C	0.751469	3.116335	0.000000
C	2.387887	1.325038	0.000000
C	3.411389	2.268367	0.000000
C	1.781635	4.052848	0.000000
H	-0.285188	3.439240	0.000000
H	2.615576	0.263624	0.000000
H	4.446208	1.936385	0.000000
H	1.544197	5.113304	0.000000
H	3.916621	4.366450	0.000000

$\theta = 171^\circ$ $E = -539.459750$ Hartree			
			
H	-0.461826	-3.388260	0.000000
C	0.593898	-3.134613	0.000000
C	3.304478	-2.464507	0.000000
C	0.972744	-1.777441	0.000000
C	1.560079	-4.137048	0.000000
C	2.917563	-3.807257	0.000000
C	2.345472	-1.455727	0.000000
H	1.253002	-5.179485	0.000000
H	3.669614	-4.591454	0.000000
H	2.642995	-0.411713	0.000000
H	4.358983	-2.201699	0.000000
C	0.000000	-0.734512	0.000000
C	-0.681817	0.274054	0.000000

C	-1.285751	1.566063	0.000000
C	-2.442708	4.127941	0.000000
C	-2.686213	1.721263	0.000000
C	-0.474705	2.719363	0.000000
C	-1.052581	3.985642	0.000000
C	-3.255520	2.991804	0.000000
H	-3.315598	0.836519	0.000000
H	0.604875	2.605500	0.000000
H	-0.415127	4.865819	0.000000
H	-4.337208	3.096181	0.000000
H	-2.889331	5.118440	0.000000

$\theta = 172^\circ$

$E = -539.460255$ Hartree

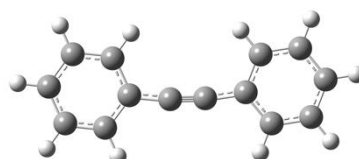


H	-3.240543	-0.982623	0.000000
C	-2.581937	-1.845807	0.000000
C	-0.872858	-4.054242	0.000000
C	-1.187346	-1.643858	0.000000
C	-3.108056	-3.134804	0.000000
C	-2.257409	-4.242986	0.000000
C	-0.338003	-2.769253	0.000000
H	-4.185613	-3.275572	0.000000
H	-2.670576	-5.247911	0.000000
H	0.737119	-2.619106	0.000000
H	-0.206319	-4.912584	0.000000
C	-0.628292	-0.332081	0.000000
C	0.000000	0.710467	0.000000
C	0.898351	1.817805	0.000000
C	2.695525	3.979259	0.000000
C	0.424905	3.145020	0.000000
C	2.290268	1.593315	0.000000
C	3.175966	2.667083	0.000000
C	1.318218	4.212868	0.000000

H	-0.646073	3.323596	0.000000
H	2.660315	0.572754	0.000000
H	4.246366	2.479217	0.000000
H	0.938691	5.231146	0.000000
H	3.390639	4.814357	0.000000

$\theta = 173^\circ$

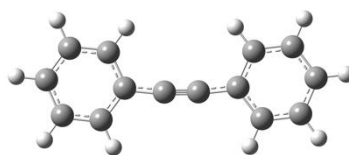
$E = -539.460727$ Hartree



H	-0.843657	3.241421	0.000000
C	0.237885	3.146004	0.000000
C	3.017823	2.881454	0.000000
C	0.812416	1.859201	0.000000
C	1.046296	4.279422	0.000000
C	2.437522	4.152652	0.000000
C	2.217443	1.742657	0.000000
H	0.589482	5.265457	0.000000
H	3.066130	5.038906	0.000000
H	2.665024	0.753628	0.000000
H	4.099517	2.776721	0.000000
C	0.000000	0.687644	0.000000
C	-0.567431	-0.389062	0.000000
C	-1.073162	-1.722050	0.000000
C	-2.042097	-4.360672	0.000000
C	-2.458998	-1.977817	0.000000
C	-0.181030	-2.813753	0.000000
C	-0.665881	-4.118472	0.000000
C	-2.934873	-3.286143	0.000000
H	-3.150448	-1.140703	0.000000
H	0.887533	-2.622289	0.000000
H	0.033319	-4.950429	0.000000
H	-4.006208	-3.468306	0.000000
H	-2.416163	-5.380795	0.000000

$\theta = 174^\circ$

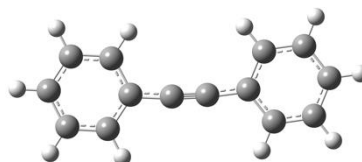
$E = -539.461151$ Hartree



H	-3.050823	-1.296626	0.000000
C	-2.326000	-2.104979	0.000000
C	-0.447162	-4.171352	0.000000
C	-0.951603	-1.793384	0.000000
C	-2.748145	-3.431617	0.000000
C	-1.812458	-4.469031	0.000000
C	-0.015893	-2.847972	0.000000
H	-3.811179	-3.657269	0.000000
H	-2.144970	-5.503456	0.000000
H	1.044015	-2.613207	0.000000
H	0.285156	-4.974292	0.000000
C	-0.502123	-0.440578	0.000000
C	0.000000	0.667931	0.000000
C	0.720331	1.898046	0.000000
C	2.162845	4.310445	0.000000
C	0.047641	3.136366	0.000000
C	2.130173	1.890822	0.000000
C	2.840013	3.088138	0.000000
C	0.765943	4.328944	0.000000
H	-1.038028	3.147680	0.000000
H	2.653082	0.939453	0.000000
H	3.926566	3.067631	0.000000
H	0.234036	5.276590	0.000000
H	2.720770	5.242811	0.000000

$\theta = 175^\circ$

$E = -539.461486$ Hartree

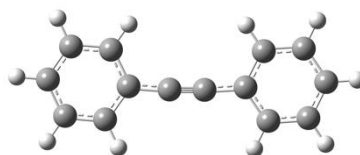


H	-2.935202	-1.458708	0.000000
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C	-2.174612	-2.233470	0.000000
C	-0.204128	-4.213021	0.000000
C	-0.815634	-1.860179	0.000000
C	-2.536188	-3.577859	0.000000
C	-1.554556	-4.571972	0.000000
C	0.166734	-2.871393	0.000000
H	-3.587909	-3.851468	0.000000
H	-1.840096	-5.620342	0.000000
H	1.214956	-2.588880	0.000000
H	0.563743	-4.982016	0.000000
C	-0.430154	-0.487937	0.000000
C	0.000000	0.650282	0.000000
C	0.617334	1.935016	0.000000
C	1.854913	4.458621	0.000000
C	-0.155783	3.113360	0.000000
C	2.022843	2.044648	0.000000
C	2.631068	3.296701	0.000000
C	0.461247	4.361338	0.000000
H	-1.238641	3.034586	0.000000
H	2.622845	1.139910	0.000000
H	3.715573	3.366388	0.000000
H	-0.147346	5.261664	0.000000
H	2.333575	5.434054	0.000000

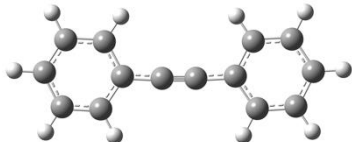
$\theta = 176^\circ$

$E = -539.461754$ Hartree



H	-1.442242	-2.898680	0.000000
C	-0.370688	-3.073375	0.000000
C	2.388786	-3.503543	0.000000
C	0.504062	-1.968326	0.000000
C	0.132919	-4.371194	0.000000
C	1.512453	-4.591931	0.000000
C	1.894316	-2.202382	0.000000

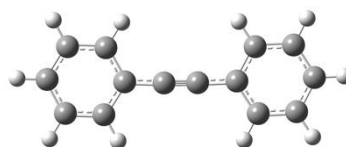
H	-0.553341	-5.213824	0.000000
H	1.902588	-5.606024	0.000000
H	2.572423	-1.354581	0.000000
H	3.462793	-3.669410	0.000000
C	0.000000	-0.635206	0.000000
C	-0.350211	0.529976	0.000000
C	-0.664365	1.920146	0.000000
C	-1.269038	4.665084	0.000000
C	-2.003355	2.360071	0.000000
C	0.366608	2.881698	0.000000
C	0.062030	4.239948	0.000000
C	-2.298450	3.720508	0.000000
H	-2.801131	1.623666	0.000000
H	1.399721	2.548103	0.000000
H	0.866792	4.970263	0.000000
H	-3.335485	4.045391	0.000000
H	-1.502521	5.726246	0.000000

$\theta = 177^\circ$ $E = -539.461983$ Hartree			
			
H	-1.640906	2.744662	0.000000
C	-0.589836	3.016746	0.000000
C	2.118959	3.697077	0.000000
C	0.382312	1.996201	0.000000
C	-0.206799	4.355098	0.000000
C	1.146879	4.700814	0.000000
C	1.745319	2.356153	0.000000
H	-0.967092	5.131561	0.000000
H	1.442812	5.746301	0.000000
H	2.498108	1.573924	0.000000
H	3.173302	3.960477	0.000000
C	0.000000	0.623329	0.000000
C	-0.266126	-0.563824	0.000000

C	-0.503833	-1.968965	0.000000
C	-0.962156	-4.742173	0.000000
C	-1.817672	-2.479290	0.000000
C	0.576584	-2.874509	0.000000
C	0.344434	-4.247066	0.000000
C	-2.040188	-3.853475	0.000000
H	-2.653289	-1.786103	0.000000
H	1.590608	-2.486753	0.000000
H	1.186738	-4.933760	0.000000
H	-3.058568	-4.232764	0.000000
H	-1.138969	-5.814235	0.000000

$\theta = 178^\circ$

$E = -539.462117$ Hartree

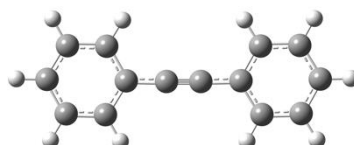


H	-2.504747	1.929200	0.000000
C	-1.635023	2.579066	0.000000
C	0.614487	4.234861	0.000000
C	-0.348704	2.002805	0.000000
C	-1.787131	3.962829	0.000000
C	-0.665322	4.795626	0.000000
C	0.776300	2.852318	0.000000
H	-2.784884	4.393469	0.000000
H	-0.787534	5.875280	0.000000
H	1.769281	2.413468	0.000000
H	1.490584	4.877879	0.000000
C	-0.183433	0.587371	0.000000
C	0.000000	-0.615295	0.000000
C	0.263608	-2.015729	0.000000
C	0.793874	-4.776073	0.000000
C	-0.792500	-2.949180	0.000000
C	1.590615	-2.491426	0.000000
C	1.848353	-3.859352	0.000000
C	-0.525114	-4.315417	0.000000

H	-1.816446	-2.588249	0.000000
H	2.407663	-1.776544	0.000000
H	2.876307	-4.211932	0.000000
H	-1.349162	-5.023893	0.000000
H	0.998873	-5.843101	0.000000

$\theta = 179^\circ$

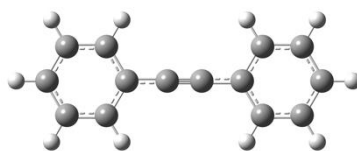
$E = -539.462197$ Hartree



H	-2.335896	-2.070279	0.000000
C	-1.431785	-2.671318	0.000000
C	0.905442	-4.201085	0.000000
C	-0.179023	-2.025272	0.000000
C	-1.507519	-4.061287	0.000000
C	-0.341624	-4.831246	0.000000
C	0.990933	-2.811694	0.000000
H	-2.480099	-4.546148	0.000000
H	-0.404474	-5.915974	0.000000
H	1.958306	-2.318866	0.000000
H	1.815548	-4.794957	0.000000
C	-0.094375	-0.602782	0.000000
C	0.000000	0.610081	0.000000
C	0.135409	2.028633	0.000000
C	0.407617	4.826134	0.000000
C	-1.002773	2.860184	0.000000
C	1.412584	2.625235	0.000000
C	1.542533	4.011139	0.000000
C	-0.863084	4.245249	0.000000
H	-1.988793	2.405758	0.000000
H	2.292382	1.989169	0.000000
H	2.533356	4.457531	0.000000
H	-1.749194	4.874361	0.000000
H	0.512870	5.907577	0.000000

$\theta = 180^\circ$

$E = -539.462244$ Hartree



C	-4.142739	-1.208440	-0.000012
C	-2.750715	-1.213334	-0.000015
C	-2.033188	-0.000036	0.000002
C	-2.750652	1.213320	0.000020
C	-4.142663	1.208505	0.000019
C	-4.843952	0.000046	0.000007
H	-4.683092	-2.151299	-0.000024
H	-2.202977	-2.150736	-0.000028
H	-2.202838	2.150676	0.000033
H	-4.682979	2.151385	0.000033
H	-5.930517	0.000088	0.000011
C	-0.608271	-0.000050	-0.000005
C	0.608204	-0.000136	-0.000028
C	2.033182	-0.000061	-0.000019
C	2.750764	-1.213352	0.000008
C	2.750624	1.213296	-0.000021
C	4.142777	-1.208422	0.000028
H	2.203042	-2.150759	0.000019
C	4.142650	1.208518	-0.000010
H	2.202813	2.150653	-0.000034
C	4.843967	0.000092	0.000014
H	4.683170	-2.151258	0.000049
H	4.682921	2.151424	-0.000014
H	5.930531	0.000142	0.000027

[S1] M. J. Frisch, G. W. Trucks, H. B. Schlegel, G. E. Scuseria, M. A. Robb, J. R. Cheeseman, G. Scalmani, V. Barone, B. Mennucci, G. A. Petersson, H. Nakatsuji, M. Caricato, X. Li, H. P. Hratchian, A. F. Izmaylov, J. Bloino, G. Zheng, J. L. Sonnenberg, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, T. Vreven, J. A. Montgomery, Jr., J. E. Peralta, F. Ogliaro, M. Bearpark, J. J. Heyd, E. Brothers, K. N. Kudin, V. N. Staroverov, T. Keith, R. Kobayashi, J. Normand, K.

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